01 Mole Concept

All of the objects around you - this book, your pen or pencil, and the things of nature such as rocks, water and plant and animal substances - constitute the matter of the universe. Each of the particular kinds of matter, such as a certain kind of paper or plastic or metal, is referred to as a material. We can define chemistry as the science of the composition and structure of materials and of the changes that material undergoes.

Because chemistry deals with all materials, it is a subject of enormous breadth. It would be difficult to exaggerate the influence of the chemistry on modern science and technology. In the section that follows, we will take a brief glimpse at basic concepts of chemistry.

Classification of Matter:

Matter: Matter is anything that has mass and occupies space.

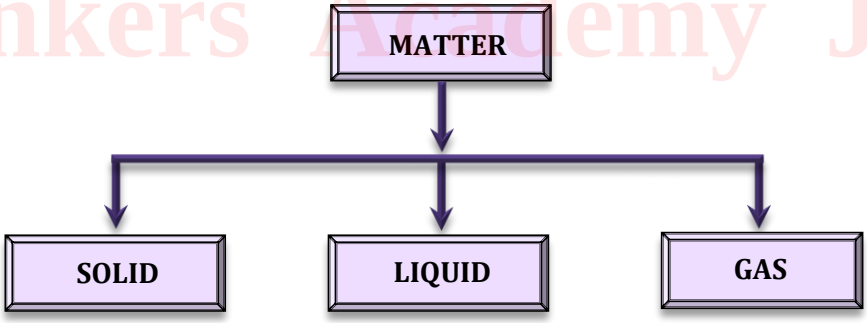
Two ways of classifying matter.

I. Physical classification

II. Chemical classification

I. Physical classification:

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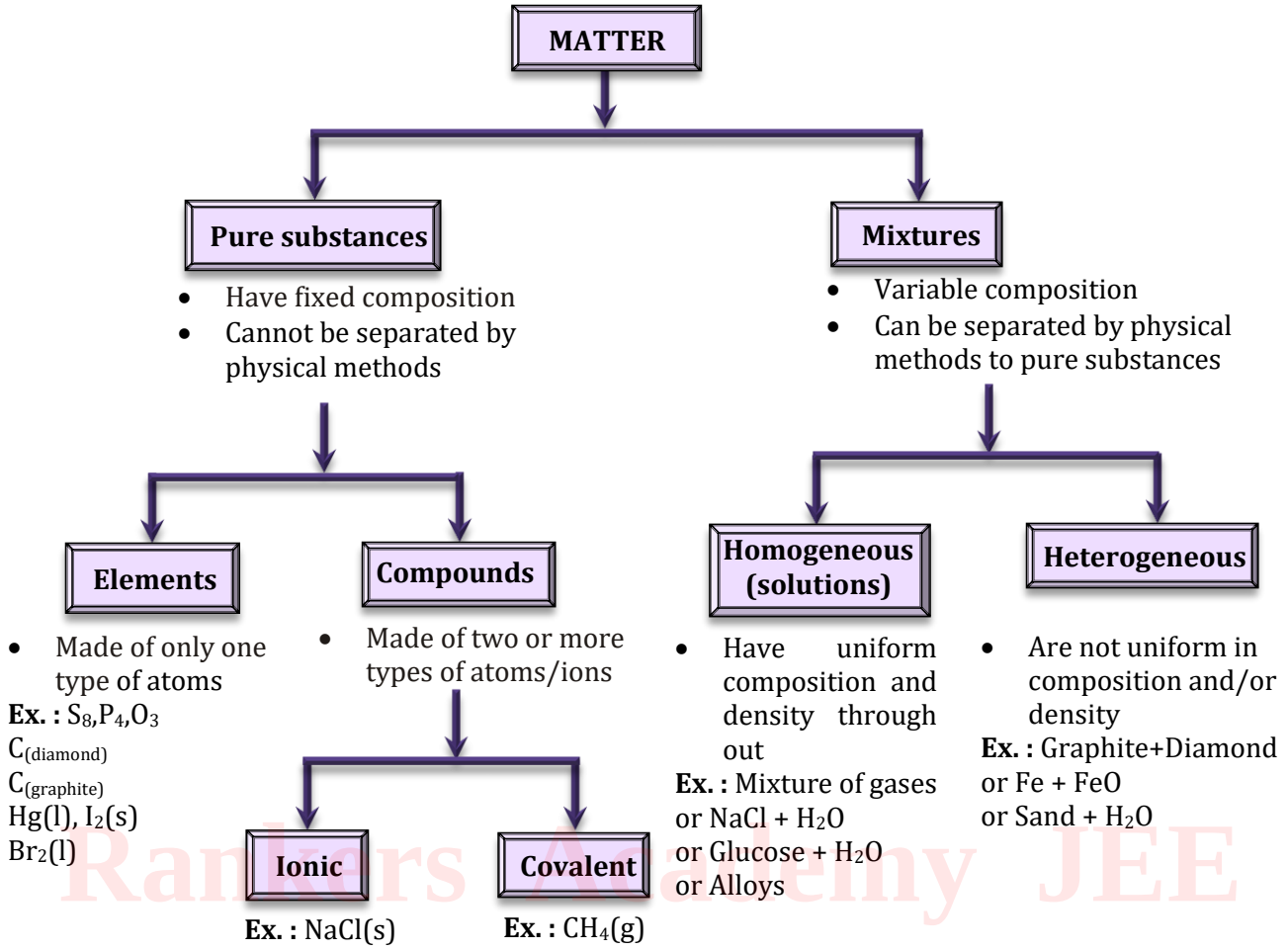
(i)	Particles are very closely packed in ordered manner.	Particles are less closely packed.	Particles are farthest apart
(ii)	No freedom of movement of particles	Particles can move around to some extent	Movement of particles is very easy and fast
(iii)	Definite shape and volume	Definite volume, indefinite shape	indefinite shape and volume
(iv)	Exists at low T and high P	Exists at intermediate P & T	Exists at high T and low P

Note: For same substance:

- Solid and Liquid co-exist at ***MELTING POINT.***
- Liquid and gas co-exist at ***BOILING POINT.***
- Solid and gas co-exist at ***SUBLIMATION POINT.***
- Solid, liquid and gas co-exist at ***TRIPLE POINT.***

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II. Chemical classification:



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Note : PHASE : It is the state of matter uniform in density and composition.

- Homogeneous mixtures have single phase while heterogeneous mixtures are multi-phase.
  Ex : NaCl + H_2O mixture has one phase
  Ex : Graphite + Diamond mixture has 2 phases.

Illustration 1

Match the column :

Column-I		Column-II	
(A)	Element	(P)	Can't be decomposed into simpler substances by physical decomposition
(B)	Compound	(Q)	Can be decomposed into simpler substances by chemical methods.
(C)	Mixture	(R)	Can't be decomposed into simpler substances by chemical methods
(D)	H_2O	(S)	Uniform composition
(E)	Fe(s) + S(s) in powdered form	(T)	Consists of one type of atoms
(F)	Sea water	(U)	Consists of 2 or more type of atoms.

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Solution

(A) → (R) & (T), (P), (S)
(B) → (Q) & (U), (S), (P)
(C) → (U), (Q)
(D) → (Q) & (U), (S), (P)
(E) → (U), (Q)
(F) → (U), (Q)

Note :

(i) Mixtures can be separated by both physical and chemical methods.
(ii) Compounds can only be broken by chemical methods.

Illustration 2

$2HgO \xrightarrow{\Delta} 2Hg + O_2\uparrow$

How many phases do you observe in the above given reaction at room temperature ?

Solution

(3)

$HgO \longrightarrow$ solid, $Hg \longrightarrow$ liquid, $O_2 \longrightarrow$ gas

Properties of Matter and their Measurement:

Chemists broadly classified properties into two kinds, namely physical properties and chemical properties in order to characterize a particular sample of matter. physical properties are those properties which can be measured or observed without changing the identity or the composition of the substance.

Chemical properties are those which can be measured only by a chemical reaction. these cannot be determined just by viewing or touching the substance. chemical properties shows the kind of reaction in which a substance can take part e.g ., acidity or basicity, combustibility, reactivity etc.

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There are many properties of matter which are capable of being measured or quantitative in nature e.g., length, area, volume etc. such quantities are measured with some fixed standard. for example, if we want to know the length of a piece of paper, we shall lay the piece of paper parallel to a centimeter scale and will count the number of marking on the scale. suppose the number of marking is 15.2, this means that the piece of paper is 15.2 times longer than one unit of measurement but it is senseless to say the paper is 15.2, until we add centimeter, a unit of length to 15.2 i.e., 15.2 cm. this presentation is appropriate and called **physical quantity**. therefore, a measured physical quantity is expressed in two parts, a **numerical coefficient** and a **unit**. either of them are meaningless without each other.

A unit is the standard of reference chosen to measure or express any physical quantity.

The International System of Units (S.I.):

SI system is a modification of metric system and has seven base units pertaining to the seven fundamental scientific quantities.

Base physical quantities and their units

Base Physical Quantity	Symbol for Quantity	Name of SI unit	Symbol for SI unit
Length	L	Metre	m
Mass	m	Kilogram	Kg
Time	t	Second	s
Electric current	I	Ampere	A
Thermodynamic temperature	t	Kelvin	K
Amount of substance	n	Mole	mol
Luminous intensity	I_v	Candela	cd

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Some specific properties of substances:

Deliquescence :

The property of certain compounds of taking up the moisture present in atmosphere and becoming wet when exposed, is known as deliquescence. These compounds are known as deliquescent. Sodium hydroxide, potassium hydroxide, anhydrous calcium chloride, anhydrous magnesium chloride, anhydrous ferric chloride, etc., are the examples of deliquescent compounds.

Hygroscopicity :

Certain compounds combine with the moisture of atmosphere and are converted into hydroxides or hydrates. Such substances are called hygroscopic. Anhydrous copper sulphate, quick lime (CaO), anhydrous sodium carbonate, etc., are of hygroscopic nature.

Efflorescence :

The property of some crystalline substances of losing their water of crystallisation on exposure and becoming powdery on the surface is called efflorescence and such salts are known as efflorescent. The examples are : Ferrous sulphate ($FeSO_4.7H_2O$), sodium carbonate ($Na_2CO_3.10H_2O$), sodium sulphate ($Na_2SO_4.10H_2O$), potash alum [$K_2SO_4.Al_2(SO_4)_3.24H_2O$], etc.

Malleability :

This property is shown by metals. When metallic solid is being beaten, it does not break but is converted into thin sheet. It is said to possess the property of malleability. Copper, gold, silver, aluminium, lead, etc., can be easily hammered into sheets. Gold is the most malleable metal.

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Ductility :

The property of metal to be drawn into wires is termed ductility. Copper, silver, gold, aluminium, iron, etc., are ductile in nature. Platinum is the most ductile metal.

Brittleness :

The solid materials which break into small pieces on hammering are called brittle. The solids of non-metals are generally brittle in nature.

Ex: Ice, Diamond etc.

Some Basic Concepts:

At the center of each atom is a "***nucleus***" that consists of two types of particles called "***protons***" and "***neutrons***" (and known collectively as "nucleons"). The number of protons is called the "***atomic number***". The total number of protons and neutrons in the nucleus of the atom is called the "***mass number***" and is determined by the type of atom (i.e. which element it is) and also by the number of neutrons present in that particular atom. In many cases, different atoms of the same element can have different numbers of neutrons in their nuclei ("nuclei" is the plural word for "nucleus"; one nucleus - many nuclei). The number of neutrons in an atom of a specific element therefore determines its "mass number".

For example, most atoms of the element phosphorus have a mass number of 31 and so may be referred to as phosphorus-31, however, some atoms of phosphorus have an extra neutron and therefore a mass number of 32. They are therefore atoms of the isotope phosphorus-32.

Although there are three types of particles that form atoms (protons, neutrons and electrons), electrons have very low mass compared with protons and neutrons so the mass of atoms is mainly due to the numbers of protons and neutrons that make-up the atom, i.e. the "mass number" of the atom.

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The atomic number of an atom is the same as the atomic number of the element of which that atom is a particle because the atomic number of an atom determines which element it is an atom of.

However, the mass number of an atom is not the same as the mass number of an element because atoms of the same element can have different masses due to having different numbers of neutrons (i.e. exist in the form of different isotopes, which therefore have slightly different physical properties e.g. boiling points).

That is why Atomic Number of an element is meaningful, but the Mass Number "of an element" is not so meaningful.

When dealing with elements (macroscopic scale) rather than atoms (individual particles), chemists use Relative Atomic Mass because "mass number" only applies to specific atoms of one particular isotope of the element.

Atom is actually made of 3 fundamental particles:

1. Electron **2.** Proton **3.** Neutron

Fundamental particle	Discovered By	Charge	Mass	$\left(\frac{\text{charge}}{\text{mass}}\right)$ (Specific Charge)
Electron ($e^{\ominus}$ or β)	J. J. Thomson	-1.6×10^{-19} coloumb -4.8×10^{-10} esu −1 Unit	9.1×10^{-31}kg 9.1×10^{-28}g 0.000544 a.m.u.	1.76×10^{8} C/g
Proton (P) **(Ionized H atom, $H^{\oplus}$)**	Rutherford	$+1.6 \times 10^{-19}$ coloumb $+4.8 \times 10^{-10}$esu 9.58×10^{4} C/g +1 Unit	1.672×10^{-27}kg 1.672×10^{-24}g 1.00727 a.m.u.	9.58×10^{4}C/g
Neutron ($_0n^1$)	James Chadwick	0	1.675×10^{-27}kg 1.675×10^{-24}g 1.00867 a.m.u.	0

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Representation of atom : $_ZX^A$

A → Mass number : (total number of protons + total number of neutrons present in an atom.)

Z → Atomic number : (total number of protons present in an atom.)

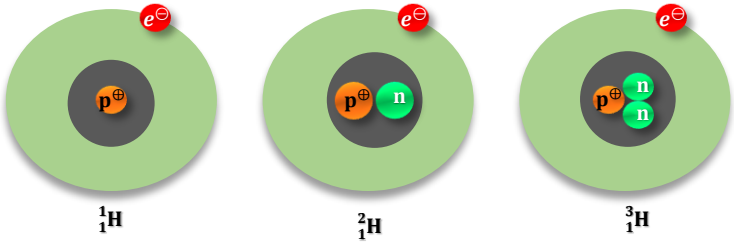
⇒ ***Isotope :*** Atoms of given element which have same atomic number but different mass number are called isotope :

e.g. $_1H^1$, $_1H^2$, $_1H^3$,

e.g. , $_8O^{16}$, $_8O^{18}$

e.g. , $_{17}Cl^{35}$, $_{17}Cl^{37}$

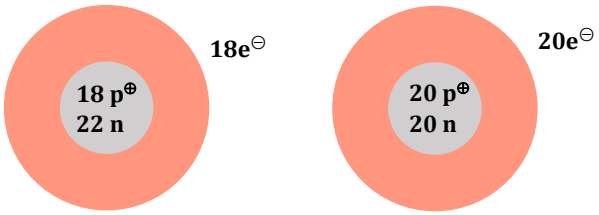
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⇒ ***Isobar :*** Atoms of different elements with the same mass number but different atomic number .

e.g. $_{18}Ar^{40}$, $_{19}K^{40}$ and $_{20}Ca^{40}$

e.g. $_{6}C^{14}$, $_{7}N^{14}$



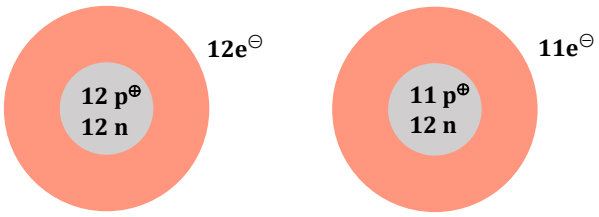
⇒ ***Isotones :*** Isotones are atoms that have the same number of neutrons but different number of protons.

e.g $_{12}Mg^{24}$, $_{11}Na^{23}$

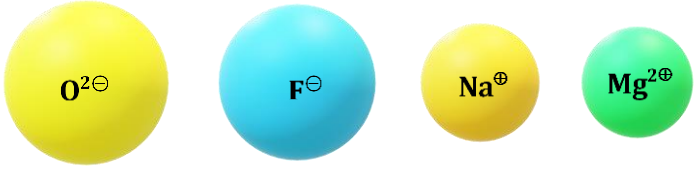
e.g. $_{9}F^{19}$, $_{8}O^{18}$

$_{15}P^{31}$, $_{16}S^{32}$

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⇒ ***Iso-electronic species :*** Species (atom, molecules or ions) having same number of electrons are called iso-electronic e.g $O^{2\ominus}$, $F^{\ominus}$, $Na^{\oplus}$, $Mg^{2\oplus}$



Note : Now a days this concept is extended to consider the same valence shell electron also.

eg. $SO_4^{2\ominus}$, $SeO_4^{2\ominus}$, $TeO_4^{2\ominus}$

eg. $ClO_4^{\ominus}$, $BrO_4^{\ominus}$, $IO_4^{\ominus}$

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⇒ ***Iso-sters :*** Species having same number of electrons & same number of atoms. eg. N_2O, CO_2 $NO_3^{\ominus}$, CO_3^{2}, $BO_3^{3\ominus}$

eg. Number of atoms = 4

Number of electrons in $NO_3^{\ominus}$ = 7 + 8 + 8 + 1 =32

Number of electrons in $CO_3^{2\ominus}$ = 6 + 8 + 8 + 8 + 2 = 32

Number of electrons in $BO_3^{3\ominus}$ = 5 + 8 + 8 + 8 + 3 = 32

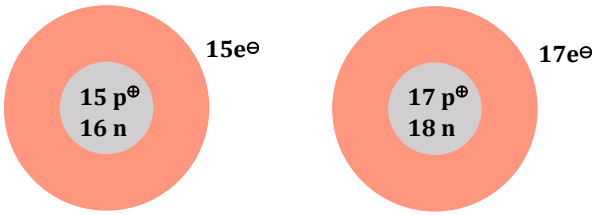
eg. CH_4, $NH_4^{\oplus}$

eg. $PO_4^{3\ominus}$, $SO_4^{2\ominus}$, $ClO_4^{\ominus}$

⇒ ***Iso-diaphers :*** Species having same difference in number of neutrons and protons or same number of excess of neutron. eg. $^{19}_{9}F^{23}$, $_{11}Na$

eg. $_{15}P^{31}$, $_{17}Cl^{35}$

eg. $_{17}Cl^{37}$, $_{21}Si^{45}$



Neutron – Proton = (n – p) = (A – 2Z). It is also known as neutron excess.

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The law of chemical combination:

Atoine Lavoisier, John Dalton and other scientists formulated certain laws concerning the composition of matter and chemical reactions. These laws are known as the law of chemical combination.

I. Law of indestructibility of matter or conservation of Mass:

- This law was proposed by ***Lavosier in 1774.***
- The experimental certification was given by Landolt.
- According to this law in all chemical changes the total mass of the system remains constant or in a chemical change mass is neither created nor destroyed. Thus, in chemical change-

  Total mass of reactant = Total mass of products

  Ex. $H_2O(s) \longrightarrow H_2O(\ell)$

  Above reaction shows the physical change and the wt. of H_2O (s) = wt. of (H_2O) (ℓ)

  In case the reacting materials are not completely consumed, the relationship will be Total masses of reactants = Total masses of product + masses of unreacted reactants
- In nuclear reactions (Mass + energy) is conserved not the mass separately.



Antoine Lavosier
(1743-1794)

Antoine-Laurent de Lavosier, the "Father of modern chemistry", was a French nobleman prominent in the histories of chemistry and biology. He named both oxygen and hydrogen and predicted silicon.

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Illustration 3

When 4.2 g $NaHCO_3$ is added to a solution of CH_3COOH weighing 10.0 g, it is observed that 2.2 g CO_2 is released into atmosphere. The residue is found to weigh 12.0 g. Show that these observations are in agreement with the law of conservation of mass.

Solution

$NaHCO_3 + CH_3COOH \longrightarrow CH_3COONa + H_2O + CO_2$

Initial mass = 4.2 + 10 = 14.2

Final mass = 12 + 2.2 = 14.2

Thus, during the course of reaction law of conservation of mass is obeyed.

II. Law of constant or definite proportion:

- This law was given by ***Joseph Louis proust. in 1799***.



Joseph Proust (1754 - 1826)

Proust was born the son of an apothecary at Angers in north-west France. He studied in Paris. He lived in poverty for some years before being awarded a pension by Louis XVIII.

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- Chemical composition of a compound remains constant whether it is obtained by any method or any source.

  Example :

  In water (H_2O), Hydrogen and Oxygen combine in 2 : 1 molar ratio, the ratio remains constant whether it is tap water, river water or sea water or produced by any chemical reaction.

Illustration 4

1.80 g of a certain metal burnt in oxygen gave 3.0 g of its oxide. 1.50 g of the same metal heated in steam gave 2.50 g of its oxide. Show that these results illustrate the law of constant proportion.

Solution

In the first sample of the oxide,

wt. of metal = 1.80 g, wt. of oxygen = (3.0 – 1.80) g = 1.2 g

$\therefore \quad \frac{\text{wt.of metal}}{\text{wt.of oxygen}} = \frac{1.80\text{g}}{1.2\text{g}} = 1.5$

In the second sample of the oxide,

wt. of metal = 1.50 g, wt. of oxygen = (2.50 – 1.50) g = 1 g

$\therefore \quad \frac{\text{wt.of metal}}{\text{wt.of oxygen}} = \frac{1.50\text{g}}{1\text{g}} = 1.5$

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Thus, in both samples of the oxide the proportions of the weights of the metal and oxygen are fixed. Hence, the results follow the law of constant proportion.

Note :

This law is not applicable in case of isotopes.

III. The law of multiple proportion:

- This law was given by Dalton in 1804.
- If two elements combine to form more than one compound, then the different masses of one element which combine with a fixed mass of the other element, bear a simple ratio to one another.

Ex. Nitrogen and oxygen combine to form five stable oxides –

N_2O	Nitrogen 28 parts	Oxygen 16 parts
N_2O_2	Nitrogen 28 parts	Oxygen 32 parts
N_2O_3	Nitrogen 28 parts	Oxygen 48 parts
N_2O_4	Nitrogen 28 parts	Oxygen 64 parts
N_2O_5	Nitrogen 28 parts	Oxygen 80 parts

The masses of oxygen which combine with same mass of nitrogen in the five compounds bear a ratio 16 : 32 : 48 : 64 : 80 or 1 : 2 : 3 : 4 : 5.

Note :

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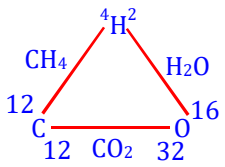
This law is not applicable in case of isotopes.

IV. Law of reciprocal proportion (or law of equivalent wt.):

This law was put forward by ***Richter in 1792***. It states as follows :

The ratio of the weights of two elements A and B which combine separately with a fixed weight of the third element C is either the same or some simple multiple of the ratio of the weights in which A and B combine directly with each other. This law may be illustrated with the help of the following example.

The elements C and O combine separately with the third element H to form CH_4 and H_2O and they combine directly with each other to form CO_2, as shown in fig.



In CH_4, 12 parts by weight of carbon combine with 4 parts by weight of hydrogen. In H_2O, 2 parts by weight of hydrogen combine with 16 parts by weights of oxygen. Thus the weight of C and O which combine with fixed weight of hydrogen (say 4 parts by weight) are 12 and 32 i.e. they are in the ratio 12 : 32 or 3 : 8.

Now in CO_2, 12 parts by weight of carbon combine directly with 32 parts by weight of oxygen i.e. they combine directly in the ratio 12 : 32 or 3 : 8 which is the same as the first ratio.

Special Note :

This law is also called as law of equivalent wt. due to each element combined in their equivalent wt. ratio.

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Example based on Law of Reciprocal proportion:

Illustration 5

Ammonia contains 82.35% of nitrogen and 17.65% of hydrogen. Water contains 88.90% of oxygen and 11.10% of hydrogen. Nitrogen trioxide contains 63.15% of oxygen and 36.85% of nitrogen. Show that these data illustrate the law of reciprocal proportions.

Solution

In NH_3, 17.65g of H combine with N = 82.35g

$\therefore$ 1 g of H combine with N = $\frac{82.35}{17.65}$ g = 4.66 g

In H_2O, 11.10 g of H combine with O = 88.90 g

$\therefore$ 1 g of H combine with O = $\frac{88.90}{11.10}$ g = 8.00 g

$\therefore$ Ratio of the weights of N and O which combine with fixed weight (=1g) of H = 4.66 : 8.00 = 1 : 1.7

In N_2O_3, ratio of weights of N and O which combine with each other = 36.85 : 63.15 = 1 : 1.7

Thus, the two ratios are the same. Hence it illustrates the law of reciprocal proportions.

V. Law of Gaseous volumes:

- This law was given by ***Gay-Lussac.*** in 1808.



Joseph Louis Gay Lussac
(1778 - 1850)

Joseph Louis Gay-Lussac also; 6 December 1778 – 9 May1850) was a French chemist and physicist. He is known mostly for two laws related to gases, and for his work on alcohol-water mixtures, which led to the degrees Gay- Lussac used to measure alcoholic beverages in many countries.

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- According to this law, gases react with each other in the simple ratio of their volumes. If products are also gases then they are also in simple ratio of volume provided that all volumes are measure at same temp. & pressure.

eg. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$
1 vol. 3vol. 2vol.

Illustration 6

For the gaseous reaction, $H_2 + Cl_2 \longrightarrow 2HCl$

If 40 ml of hydrogen completely reacts with chlorine then find out the required volume of chlorine and volume of produced HCl ?

Solution

According to Gay Lussac's Law :

$H_2 + Cl_2 \longrightarrow 2HCl$

$\because$ 1 ml of H_2 will react will 1 ml of Cl_2 and 2 ml of HCl will produce.

$\therefore$ 40 ml of H_2 will react with 40 ml of Cl_2 and 80 ml of HCl will produce.

required vol. of Cl_2 = 40 ml, produced vol. of HCl = 80 ml

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VI. Berzelious Hypothesis and Avo-Gadro's Hypothesis:

(A) Berzelious Hypothesis : Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of atoms.



Berzelius
(1779 - 1848)
Jöns Jacob Berzelius was a Swedish chemist. He worked out the modern technique of chemical formula notation, and is together with John Dalton, Antoine Lavoisier, and Robert Boyle considered a father of modern chemistry.

The above statements was incorrect and later it was modified by Avo-Gadro.

(B) Avo-Gadro's Hypothesis : Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules.



Lorenzo Romano Amedeo Carlo Avogadro Di Quareqa edi Carreto
(1776-1856)
Italian mathematical physicist. He practiced law for many years before he became interested in science. His most famous work, now known as Avogadro's law, was largely ignored during his lifetime, although it became the basis for determining atomic masses in the late nineteenth century.

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Application of Avogadro's hypothesis:

(a) In finding the atomicity

(b) Relation between molecular weight and vapour density

- Vapour density $= \frac{\text{density of gas}}{\text{density of } H_2}$ (constant P, T)
- Molecular weight = 2 × vapour density
- Density of H_2 = 0.000089 gm/cm^3 at 0°C & 1 atm

  ≈ 0.089 gm/lit.

(c) Relation between molecular weight and volume.

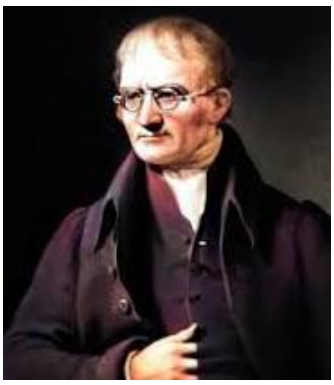
- 1 molecular weight = 22.4 lit. volume of gas at 0°C & 1 atm
- Weight of 1 gm mole gas = weight of 22.4 lit gas at 0°C & 1 atm
- Gram molecular volume = 22.4 lit. at 0°C & 1 atm

(d) Finding the molecular formula of gas.

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Dalton's Atomic Theory:

Ancient Indian and Greek philosophers have always wondered about the unknown and unseen form of matter. The idea of divisibility of matter was considered long back in India, around 500 BC. An Indian philosopher ***Maharishi Kanad***, postulated that if we go on dividing matter (padarth), we shall get smaller and smaller particles. Ultimately, a time will come when we shall come across the smallest particle beyond which further division will not be possible. He named these particles Parmanu. Another Indian philosopher, Pakudha Katyayama, elaborated this doctrine and said that these particles normally exist in a combined form which gives us various forms of matter. Around the same era, the Greek philosopher Democritus expressed the belief that all matter consists of very small, indivisible particles, which he named ***atomos*** (meaning uncuttable or indivisible).



John Dalton (1766 - 1844)

An Englishman, began teaching at a Quaker school when he was 12. His fascination with science included an intense interest in meteorology (he kept careful daily weather records for 46 years), which led to an interest in the gases of the air and their ultimate components, atom. Dalton is best known for his atomic theory, in which he postulated that the fundamental differences among atoms are their masses. He was the first to prepare a table of relative atomic weight.

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Although Democritus' ideal was not accepted by many of his contemporaries (notably Plato and Aristotle), somehow it endured. Experimental evidence from early scientific investigations provided support for the notion of "atomism" and gradually gave rise to the modern definitions of elements and compounds. It was in ***1808, John Dalton***, formulated a precise definition of the indivisible building blocks of matter that we call atoms. Dalton's work marked the beginning of the modern era of chemistry. The hypothesis about the nature of matter on which Dalton's atomic theory is based can be summarized as follows :

(i) Elements are composed of extremely small particles called atoms.

(ii) All atoms of a given element are identical, having the same size, mass and chemical properties. The atoms of one element are different from the atoms of all other elements.

(iii) Compounds are composed of atoms of more than one element. In any compound, the ratio of the numbers of atoms of any two of the elements present is either an integer or a simple fraction.

(iv) A chemical reaction involves only the separation, combination or rearrangement of atoms ; it does not result in their creation or destruction.

Limitations of Dalton's Atomic Theory:

According to Dalton's atomic theory, an atom is the ultimate, discrete and indivisible particle of matter. Later researches proved that Dalton's atomic theory was not wholly correct.

Dalton's atomic theory suffered from the following drawbacks:

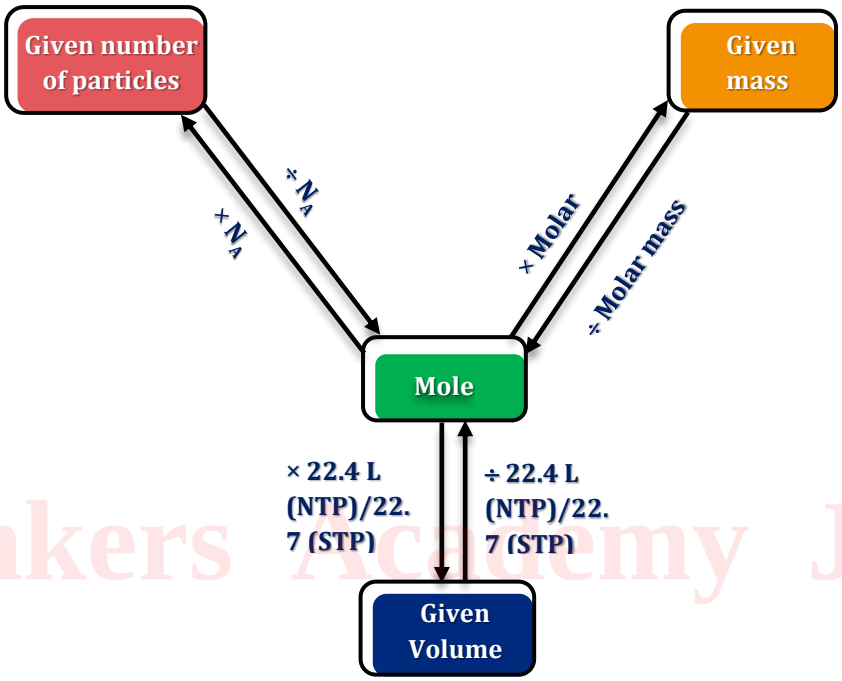
(i) Atoms of the same or different types have a strong tendency to combine together to form a new 'group of atoms'. For example, hydrogen, nitrogen, oxygen gases exist in nature as 'group of two atoms'. This indicates that the smallest unit capable of independent existence is not an atom, but a 'group of atoms'.

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(ii) With the discovery of sub-atomic particles, e.g., electrons, neutrons and protons, the atom can no longer be considered indivisible.

(iii) Discovery of isotopes indicated that all atoms of the same element are not perfectly identical. At least, they differ in their masses. Atoms of the same element having different masses are called isotopes. Dalton's atomic theory could not explain why certain substances, all containing atoms of the same element, should differ in their properties. For example, charcoal, graphite and diamond all are made up of only Carbon-atoms, but still their properties are quite different.

Introduction to Mole:



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Atoms and molecules are extremely small in size and their numbers in even a small amount of any substance is really very large. To handle such large numbers, a unit of similar magnitude is required. The 14th Geneva conference on weight and measures adopted mole as a **seventh basic SI unit of the amount of a substance.** Mole concept is essential tool for the fundamental study of chemical calculations. This concept is simple but its application requires a thorough practice. There are many ways of measuring the amount of substance, weight and volume being the most common, but basic unit of chemistry is the atom or a molecule and measuring the number of molecule is more important.

Definition of mole and molar mass:

- A mole is the amount of a substance that contains as many entities (Atoms, Molecules, Ions or any other particles) as there are atoms in exactly 12 g of C-12 isotope.
- A mole of a substance contains Avogadro's number (6.022×10^{23}) of molecules (or formula units). The term mole, like a dozen or a gross, thus refers to a particular number of things. A dozen eggs equals 12 eggs, a gross of pencils equals 144 pencils, and a mole of ethanol equal 6.022×10^{23} ethanol molecules.
- The ***molar mass*** of a substance is the mass of one mole of the substance. Carbon-12 has a molar mass of exactly 12 g/mol, by definition.

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Methods to calculate moles:

(i) If number of particles (molecules or atoms) is given then number of mole

$$\text{mole} = \frac{\text{Given no. of molecule / atom}}{N_A}$$

(ii) If mass is given then, number of mole = $\frac{\text{Given mass of substance (in gm)}}{\text{GAM / GMM}}$

(iii) If volume of gas is given then, mole = $\frac{\text{Volume of gas at STP (in lit.)}}{22.7\,\text{L}}$

(Standard molar volume is the volume occupies by 1 mole of any gas at STP, which is equal to 22.7 L)

- 1 gram-atom = 1 mole atoms = N_A atoms
- 1 gram-molecule = 1 mole molecules = N_A molecules
- 1 gram-ion = 1 mole ions = N_A ions

Illustration 7

Calculate the atomic mass (average) of chlorine using the following data :

	% Natural Abundance	Molar Mass
^{35}Cl	75.77	34.9689
^{37}Cl	24.23	36.9659

Solution

$$(\text{At. wt.})_{avg} = \frac{M_1x_1 + M_2x_2}{x_1 + x_2}$$

$$= \frac{34.9689 \times 75.77 + 36.9659 \times 24.23}{100} = 35.45$$

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Illustration 8

Use the data given in the following table to calculate the molar mass of naturally occurring argon isotopes :

Isotope	Isotopic molar mass	Abundance
^{36}Ar	35.96755 g mol^{-1}	0.337%
^{38}Ar	37.96272 g mol^{-1}	0.063%
^{40}Ar	39.9624 g mol^{-1}	99.600%

Solution

$$(\text{At wt.})_{avg} = \frac{M_1x_1 + M_2x_2 + M_3x_3}{x_1 + x_2 + x_3}$$

$$= \frac{35.96755 \times 0.337 + 37.96272 \times 0.063 + 39.9624 \times 99.6}{100}$$

$$= 39.94$$

Atomic and molecular masses:

Different types of Atomic Masses:

The mass of an atom depends on the number of electrons, protons, and neutrons it contains. Knowledge of an atom's mass is important in laboratory work. But atoms are extremely small particles - even the smallest speck of dust that our unaided eyes can detect contains as many as 1×10^{16} atoms, Clearly, we cannot weigh a single atom, but it is possible to determine the mass of one atom relative to another experimentally. The first step is to assign a value to the mass of one atom of a given element so that it can be used as a standard.

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Relative atomic mass:

Hydrogen, being lightest atom was arbitrarily assigned a mass of 1 (without any units) and other elements were assigned masses relative to it. However, the present system of atomic masses is based on carbon - 12 as the standard and has been agreed upon in 1961. Here, Carbon - 12 is one of the isotopes of carbon and can be represented as ^{12}C. In this system, ^{12}C is assigned a mass of exactly 12 atomic mass unit **(a.m.u.)** and masses of all other atoms are given relative to this standard.

It is defined as the number which indicates how many times the mass of one atom of an element is heavier in comparison to $1/12^{th}$ part of the mass of one atom of C-12.

$$\textbf{Relative atomic mass (of elements)} = \frac{\textbf{mass of one atom of an element}}{\frac{1}{12}\textbf{[mass of C-12 atom]}}$$

$$= \frac{\textbf{Mass of one atom of an element}}{\textbf{1 amu}}$$

Atomic mass unit (a.m.u. or u):

The quantity $1/12^{th}$ mass of an atom of C^{12} is known as atomic mass unit.

Since mass of 1 atom of C - 12 = 1.9924×10^{-23} g

$\therefore$ $1/12^{th}$ part of the mass of 1 atom = $\frac{1.9924\times10^{-23}\text{ g}}{12} = 1.67 \times 10^{-24}$ g

$$\Rightarrow 1 \text{ a.m.u.} = \frac{1}{6.022\times10^{23}} \text{ g}$$

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It may be noted that the atomic masses as obtained above are the relative atomic masses and not the actual masses of the atoms. These masses on the atomic mass scale are expressed in terms of atomic mass units (abbreviated as a.m.u.). Today, 'a.m.u.' has been replaced by 'u' which is known as unified mass.

One atomic mass unit (a.m.u.) is equal to $1/12^{th}$ of the mass of an atom of carbon - 12 isotope.

Thus, the atomic mass of hydrogen is 1.008 a.m.u. while that of oxygen is 15.9994 a.m.u. (or taken as 16 a.m.u.).

Gram atomic mass or mass of 1 gram atom:

When numerical value of atomic mass of an element is expressed in grams then the value becomes gram atomic mass or GAM.

gram atomic mass (GAM) = mass of 1 **gram atom** = mass of 1 **mole atoms**

= mass of N_A atoms = mass of 6.022×10^{23} atoms.

Ex. GAM of oxygen = mass of 1 **g atom** of oxygen = mass of 1 **mol atoms** of oxygen.

= mass of N_A atoms of oxygen.

$$= \left(\frac{16}{N_A}\text{g}\right)\times N_A = 16 \text{ g}$$

Ex. Mass of one atom of Oxygen = 16 a.m.u. or $16 \times 1.67 \times 10^{-24}$ g

Mass of N_A atoms of Oxygen = $16 \times 1.67 \times 10^{-24} \times 6.022 \times 10^{-23}$ g = 16 g

Now see the table given below and understand the definition given before.

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Element	R.A.M. (Relative Atomic Mass)	Atomic mass (mass of one atom)	Gram Atomic mass/ weight
N	14	14 a.m.u.	14 gm
He	4	4 a.m.u.	4 gm
C	12	12 a.m.u.	12 gm

Average atomic mass:

If an element exists in different isotopic forms (or allotropic forms) having relative abundance $X_1\%$, $X_2\%$ ..... $X_n\%$, with relative atomic masses M_1, M_2 ..... M_n respectively then,

$$\text{Avg. Atomic mass of element} = \frac{X_1}{100}(M_1)+\frac{X_2}{100}(M_2)+.....+\frac{X_n}{100}(M_n) = \sum_{i=1\text{ to }n}\frac{X_i}{100}(M_i)$$

Relative Molecular mass:

The number which indicates how many times the mass of one molecule of a substance is heavier in comparison to 1/12th part of the mass of an atom of C–12.

OR

The molecular mass of a substance is the sum of atomic masses of the elements present in a molecule. It is obtained by multiplying the atomic mass of each element by the number of its atoms and adding them together.

Ex. molecular mass of oxygen (O_2) = 32 a.m.u.
molecular mass of (O_3) = 48 a.m.u.
molecular mass of HCl = 1 + 35.5 = 36.5 a.m.u.
molecular mass of H_2SO_4 = 2 + 32 + 64 = 98 a.m.u.

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Gram molecular mass (mass of 1 gram molecule):

When numerical value of molecular mass of the substance is expressed in grams then the value becomes gram molecular mass or GMM.

gram molecular mass (GMM) = mass of 1 **gram molecule** = mass of 1 **mole molecules**
= mass of N_A molecules = mass of 6.022×10^{23} molecules

Ex. GMM of H_2SO_4 = mass of 1 **gram molecule** of H_2SO_4
= mass of 1 **mole molecules** of H_2SO_4
= mass of N_A molecules of H_2SO_4
$= \left(\frac{98}{N_A}\text{g}\right)\times N_A = 98\text{ g}$

Ex. Molecular Mass of N_2 = 28 a.m.u. = $28 \times 1.67 \times 10^{-28}$ g

Mass of N_A molecules of N_2 = $28 \times 1.67 \times 10^{-24} \times 6.022 \times 10^{23}$ g = 28 g

Thus, molecular mass can be defined as the absolute mass in grams of 6.022×10^{23} molecules of a substance.

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Average molecular mass of gas mixture:

$$= \frac{\text{Total mass of mixture of gases}}{\text{Total mole}}$$

Ex. $M_{avg} = \frac{M_A n_A + M_B n_B + M_C n_C}{n_a + n_B + n_C}$

Where n_A, n_B, $n_C \rightarrow$ moles of gases A, B, C

M_A, M_B, $M_C \rightarrow$ molecular mass of gases A, B, C

$\Rightarrow M_{avg} = M_A X_A + M_B X_B + M_C X_C$

$\Rightarrow M_{avg} = \sum M_i X_i$

where X_i = mole fraction of ith gas, M_i = molecular mass of ith gas.

Where X_A, X_B & X_C are the mole fraction of gases A, B & C.

Illustration 9

Find :

(i) No. of moles in 10^{20} atoms of Cu.

(ii) Mass of 200 $^{16}_{8}O$ atoms in a.m.u.

(iii) Mass of 100 atoms of $^{14}_{7}N$ in gm.

(iv) No. of molecules & atoms in 54 gm H_2O.

(v) No. of atoms in 88 gm CO_2.

Solution

(i) $\frac{10^{20}}{N_A}$ moles

(ii) Mass of 1 $^{16}_{8}O$ atom = 16 a.m.u.

Mass of 200 $^{16}_{8}O$ atom = 16 × 200 = 3200 a.m.u.

(iii) Mass of 1 $^{14}_{7}N$ atom = 14 a.m.u. = $14 \times 1.66 \times 10^{-24}$ g

Mass of 100 $^{14}_{7}N$ atom = $1400 \times 1.66 \times 10^{-24}$ g

(iv) Molar mass of H_2O = 18 g mol^{-1}

Moles of H_2O molecule = $\frac{54}{18} = 3$

no. of molecules in $H_2O = 3N_A$

no. of atoms in $H_2O = 3 \times 3N_A = 9N_A$

(v) Moles of CO_2 = $\frac{88}{44} = 2$ (Molar mass of CO_2 = 44)

No. of atoms in $CO_2 = 6N_A$

Illustration 10

In three moles of ethane (C_2H_6). Calculate the following :

(i) Number of moles of carbon atoms.

(ii) Number of moles of hydrogen atoms.

(iii) Number of molecules of ethane.

Solution

(i) Gram atoms of C = 3 × 2 = 6

(ii) Gram atoms of Hydrogen = 3 × 6 = 18

(iii) Number of molecules of $C_2H_6 = 3\ N_A$.

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Illustration 11

Calculate the number of atoms in each of the following :

(i) 52 moles of Ar (ii) 52 u of He (iii) 52 g of He.

Solution

(i) Number of atoms of Ar = $52N_A$

(ii) Wt. of He = $52 \times 1.66 \times 10^{-24}$ gm.

Moles of He = $\frac{52}{4} \times 1.66 \times 10^{-24}$

Number of atoms of He = $\frac{52 \times 1.66 \times 10^{-24}}{4} \times N_A = 13$ atoms

(iii) Moles of He = $\frac{52}{4} = 13\,\text{mol}$

Number of atoms of He = 13 N_A.

Illustration 12

What will be the mass of one ^{12}C atom in g ?

Solution

Wt. of 1 atom of C = 1.66×10^{-24} gm. × (At wt.)

= 1.66×10^{-24} gm × 12 = 19.92×10^{-24} gm

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Illustration 13

Calculate mass of O atoms in 6 gm CH_3COOH ?

Solution

Molar mass of CH_3COOH = 60

Moles of CH_3COOH = $\frac{6}{60} = 0.1$

1 mole CH_3COOH has 2 mol O atoms

0.1 mole CH_3COOH has 0.2 mol O atoms

Mass of oxygen = moles × Atomic mass = 0.2 × 16 = 3.2 gm.

Illustration 14

How many sucrose molecules ($C_{12}H_{22}O_{11}$) are present in 3.42 g sucrose ?

Solution

Molar mass of sucrose = 342

Moles of sucrose = $\frac{3.42}{342} = 0.01$

Molecules of sucrose = 0.01 N_A = 6.022×10^{21}

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Illustration 15

Calculate mass of water present in 499 gm $CuSO_4.5H_2O$?

(Atomic mass – Cu = 63.5, S = 32, O = 16, H = 1)

Solution

Molar mass of $CuSO_4.5H_2O$ = 249.5

moles of $CuSO_4.5H_2O = \frac{499}{249.5} = 2$

Each mole $CuSO_4.5H_2O$ has 5 mol H_2O

2 mole $CuSO_4.5H_2O$ has 10 mol H_2O

Mass of 10 mol H_2O = 10 × 18 = 180 gm.

Illustration 16

Find mass of 12.046×10^{23} atoms $^{12}C_6$ sample ?

Solution

Mass of $^{12}C_6$ atom = $12 \times 1.66 \times 10^{-24}$ g

Mass of 12.046×10^{23} atoms = $12 \times 1.66 \times 10^{-24} \times 12.046 \times 10^{23}$ = 24 gm.

Illustration 17

What is no. of O_2 molecules in 3.2×10^{-15} g sample of oxygen ?

Solution

Moles of $O_2 = \frac{3.2 \times 10^{-15}}{32} = 10^{-16}$; Molecules of $O_2 = 10^{-16} N_A = 6.022 \times 10^7$

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Illustration 18

Find no. of protons in 180 ml H_2O. Density of water = 1 gm/ml.

Solution

Mass of water = density × volume = 180 g

Moles of water $= \frac{180}{18} = 10$

1 mol water has 10 mol protons

10 mol water has 100 mol protons

10 mol water has 100 N_A protons

10 mol water has 6.022×10^{25} protons

Illustration 19

What mass of $Na_2SO_4.7H_2O$ contains exactly 6.022×10^{22} atoms of oxygen ?

Solution

Molar mass of $Na_2SO_4.7H_2O$ = 275 gm.

1 mole $Na_2SO_4.7H_2O$ has 11 mol O-atoms.

$\Rightarrow$ 11 N_A O – atoms are in 275 g $Na_2SO_4.7H_2O$

$\Rightarrow$ 6.022×10^{22} O – atoms are in $= \frac{275}{11 \times 6.022 \times 10^{23}} \times 6.022 \times 10^{22}$ g = 2.5 g

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Illustration 20

Oxygen exists as three isotopes $^{16}_{8}O, ^{17}_{8}O, ^{18}_{8}O$ with relative abundance 90%, 7% and 3% respectively. What is average atomic mass of oxygen ?

Solution

$M_{avg.} = \sum x_i M_i = 0.9 \times 16 + 0.07 \times 17 + 0.03 \times 18 = 16.13$ a.m.u. or 16.13 g/mol

Illustration 21

If an elements M has $M_{avg.} = 51.7$ find relative abundances of ^{50}M and ^{52}M isotopes in nature ?

[Assume M exists in only two allotropic forms in nature]

Solution

Let ^{50}M isotope be x % and ^{52}M isotope be (100 – x) % in nature

$\Rightarrow$ $51.7 = M_{avg.} = \frac{x}{100}(50) + \frac{100-x}{100}(52) \Rightarrow x = 15\% \Rightarrow 15\%\ ^{50}M$ and $85\%\ ^{52}M$ in nature.

Ideal Gas Equation:

(iv) Under any condition of temperature and pressure, moles of gases may be calculated using IDEAL GAS EQUATION : PV = nRT,

where, R = Universal Gas Constant

= 0.082 L-atm/K-mol

= 8.314 J/K-mol

= 2 cal/K-mol

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Units of pressure and their relation:

1 atm = 76 cm Hg

= 760 mm Hg

= 760 torr (1 torr = 1 mm Hg)

= $1.01325\ 10^6$ dyne/cm^2

= $1.01325\ 10^5$ N/m^2 or Pa

= 1.01325 bar (1 bar = 10^5 Pa)

1 bar = 75 cm Hg

Units of Volume and their relation:

1 ml = 1 cm^3 = 1 c.c.

1 Litre = 1000 ml = 1 dm^3

1 m^3 = 1000 L

Units of Temperature and their relation:

T = 273.15 + t

where, T = Absolute temperature (in Kelvin) and t = temperature in 0° C

(Normally, we take 273K in calculation)

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(v) Sometimes gas is collected over water. In this case, the measured pressure is sum of pressure of gas and the vapour pressure of water (also called Aqueous Tension). In order to calculate moles of gas, the vapour pressure of water should be deducted from the measured pressure.

If volume of gas is given then, mole

$$= \frac{\text{Volume of gas at STP}}{22.7\,\text{L}} = \frac{\text{Volume of gas at } 0^{\circ}\text{C and 1 atm}}{22.4\,\text{L}}$$

(Standard molar volume is the volume occupied by 1 mole of an ideal gas at STP (Standard temperature and pressure which is 273.15 K & 1 bar respectively), which is equal to 22.7 L).

1 mole of an ideal gas occupy 22.4 L at 0° C and 1 atm.

Illustration 22

Calculate the number of g-molecules (mole of molecules) in the following :

(i) 3.2 gm CH_4

(ii) 70 gm nitrogen

(iii) 4.5×10^{24} molecules of ozone

(iv) 2.4×10^{21} atoms of hydrogen

(v) 11.2 L ideal gas at 0° C and 1 atm

(vi) 4.54 ml SO_3 gas at STP

(vii) 8.21 L C_2H_6 gas at 400K and 2atm

(viii) 164.2 ml He gas at 27ºC and 570 torr [$N_A = 6 \times 10^{23}$]

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Solution

(i) 3.2 gram CH_4

number of moles (CH_4) = $\frac{w}{M} = \frac{3.2}{16}$ = 0.2 moles

(ii) 70 gram N_2

Number of moles = $\frac{w}{M} = \frac{70}{28} = 2.5$

(iii) 4.5×10^{24} molecules of O_3

Number of moles = $\frac{\text{no. of molecules}}{N_A} = \frac{4.5\times10^{24}}{6\times10^{23}} = 7.5$

(iv) 2.4×10^{21} atoms of hydrogen

Number of gram molecules of H_2 = $\frac{\text{no. of molecules}}{N_A} = \frac{2.4\times10^{21}}{2\times6\times10^{23}} = 0.002$

(v) 11.2 litre ideal gas at 0° C and 1 atm

Number of moles = $\frac{\text{Volume at } 0^{\circ}\text{C \& 1 atm}}{22.4\,\text{litre}} = \frac{11.2}{22.4} = 0.5$

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(vi) 4.54 ml SO_3 gas at STP

$$\text{Number of moles} = \frac{V_{STP}(ml)}{22700ml} = \frac{4.54}{22700} = 2 \times 10^{-4}$$

(vii) 8.21 litre C_2H_6 at 400 K and 2 litre

$$n = \frac{PV}{R.T} = \frac{2 \times 8.21}{0.0821 \times 400} = 0.5$$

(viii) 164.2 ml He gas at 27ºC and 570 torr

$$n = \frac{PV}{RT} = \left(\frac{570}{760}atm\right) \times \frac{164.2 \times 10^{-3} litre}{0.0821 \times 300} = 0.005$$

Illustration 23

What is number of atoms and molecules in 112 L of $O_3(g)$ at 0°C and 1atm ?

Solution

$$\text{Moles of molecules} = \frac{112}{22.4} = 5$$

Moles of atoms = 5 × 3 = 15

No. of molecules = 5 N_A

No. of atoms = 15N_A.

Density:

It is of two types.

I. Absolute density

II. Relative density

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For liquids and solids :

$$\text{Absolute density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{Relative density or specific gravity} = \frac{\text{density of the substance}}{\text{density of water at } 4°C\ (1\,gm\,ml^{-1})}$$

For gases :

$$\text{Absolute density} = \frac{\text{mass}}{\text{volume}} = \frac{PM}{RT}$$

where P is pressure of gas, M = molar mass of gas, R is the gas constant, T is absolute temperature.

Vapour Density :

Vapour density is defined as the density of the gas with respect to hydrogen gas at the same temperature and pressure.

$$\text{Vapour density} = \frac{d_{gas}}{d_{H_2}} = \frac{PM_{gas}/RT}{PM_{H_2}/RT}$$

$$\text{V.D.} = \frac{M_{gas}}{M_{H_2}} = \frac{M_{gas}}{2} \Rightarrow \boxed{M_{gas} = 2 \times \text{V.D.}}$$

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Illustration 24

A gaseous mixture of H_2 and NH_3 gas contains 68 mass % of NH_3. The vapour density of the mixture is –

Solution

No. of moles of NH_3 in 100g mixture = $\frac{68}{17}=4$

No. of moles of H_2 in 100g mixture = $\frac{32}{2}=16$

$M_{average}=\frac{\text{Total mass}}{\text{Total moles}}=\frac{100}{4+16}=5$ V D = $\frac{5}{2}$ = 2.5

Percentage Composition:

Here we are going to find out the percentage of each element in the compound by knowing the molecular formula of compound.

We know that according to law of definite proportions any sample of a pure compound always possesses constant ratio of their combining elements.

Example :

Every molecule of ammonia always has formula NH_3 irrespective of method of preparation or sources. i.e. 1 mole of ammonia always contains 1 mol of N and 3 moles of H. In other words, 17 g of NH_3 always contains 14 g of N and 3 g of H. Now find out % of each element in the compound.

Mass % of N in $NH_3=\frac{\text{Mass of N in 1 mol } NH_3}{\text{Mass of 1 mol of } NH_3}\times 100=\frac{14g}{17}\times 100=82.35\ \%$

Mass % of H in $NH_3=\frac{\text{Mass of H is 1 mol } NH_3}{\text{Mass of 1 mol of } NH_3}\times 100=\frac{3}{17}\times 100=17.65\%$

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Percentage composition of a compound refers to the amount of various constituent elements present per 100 parts by mass of the compound. It can be calculated from the formula of the compound by using the following relation :

$$\text{Percentage mass of an element}=\frac{(\text{No. of atoms of element})\times(\text{At. mass of element})\times 100}{\text{Molecular mass of substance}}$$

In case of minerals or ores the mass percentage of a particular constituent species in a formula can also be calculated using the similar relation as follows.

$$=\frac{[\text{Total mass of constituent species in one formula unit}]\times 100}{\text{Formula mass}}$$

For example, the percentage of iron Fe and FeO can be calculated in magnetite (Fe_3O_4) as follows :

Formula mass of Fe_3O_4 = 3 × 56 + 4 × 16 = 232

Atomic mass of Iron Fe = 56

Formula mass of FeO = 56 + 16 = 72

Therefore, percentage of $Fe=\frac{3\times 56\times 100}{232}$ = 72.4 % and percentage of $FeO=\frac{72\times 100}{232}=31\ \%$

Note :

- Ratio of mass of two elements in a compound is a constant (independent of amount of compound).
- Mass % of an element in a compound is a constant.

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Ex.

In $KMnO_4$

with respect to K $\rightarrow \frac{M_K}{M_{KMnO_4}} = \frac{m_K}{m_{KMnO_4}}$ M $\rightarrow$ Molar mass

with respect to O $\rightarrow \frac{4M_O}{M_{KMnO_4}} = \frac{m_O}{m_{KMnO_4}}$ m $\rightarrow$ Mass or mass % of element present in compound

Illustration 25

How many grams of $KClO_4$ contain 40 gm 'O'.

Solution

$\frac{40}{x} = \frac{64}{35.5+39+64} \Rightarrow x = 86.5$ g

Illustration 26

Calculate mass of Cu in 3.67×10^3 g $CuFeS_2$? (Atomic mass Cu = 63.5, Fe = 56, S = 32)

Solution

Molar mass of $CuFeS_2$ = 183.5

$\because \quad \frac{w_{Cu}}{w_{CuFeS_2}}$ = constant

$\therefore \quad \frac{63.5}{183.5} = \frac{w_{Cu}}{3.67\times10^3} \Rightarrow w_{Cu} = 1270$ g.

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Illustration 27

What mass of H_2SO_4 contains 32 g oxygen ?

Solution

$\frac{w_O}{w_{H_2SO_4}}$ = constant $\Rightarrow \frac{64}{98} = \frac{32}{w_{H_2SO_4}} \Rightarrow w_{H_2SO_4} = 49$ g.

Illustration 28

A sample of $MgSO_4$ has 6.022×10^{20} O atoms. What is mass of Mg in sample ?

Solution

In $MgSO_4$; $\frac{\text{mass of O}}{\text{mass of Mg}} = \frac{16\times4}{24} = \frac{8}{3}$ = constant

Mass of 6.022×10^{20} O-atoms $= \frac{6.022\times10^{20}}{6.022\times10^{23}}\times16 = 16 \times 10^{-3}$ g

$\Rightarrow \quad \frac{8}{3} = \frac{16\times10^{-3}}{\text{mass of Mg}} \Rightarrow$ Mass of Mg $= 6 \times 10^{-3}$ g

Illustration 29

If mass % of oxygen in monovalent metal 'M' carbonate (M_2CO_3) is 48%, find atomic mass of metal ?

Solution

Let A be atomic mass of M

Then $\frac{3\times16}{2A+12+3(16)} \times 100 = 48 \Rightarrow A = 20$

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Illustration 30

Find value of X if 36.6 g $BaCl_2.XH_2O$ on strong heating loses 5.4 g moisture ?
[At. mass Ba = 137, Cl = 35.5]

Solution

5.4 g H_2O is in 36.6 g $BaCl_2.XH_2O$

$$\frac{W_{H_2O}}{W_{BaCl_2\times H_2O}} = \text{const.} = \frac{5.4}{36.6} = \frac{18X}{137+71+18X} \Rightarrow X = 2$$

Illustration 31

0.492 g sample of haemoglobin has 0.34% by mass of Fe. If each molecule of haemoglobin has 4 Fe atoms find its molecular mass ?

Solution

Let M = Molecular mass of haemoglobin $\Rightarrow \frac{0.34}{100} \times M = 4 \times 56$

M = 65882.3 a.m.u..

Empirical and Molecular Formula:

We have just seen that knowing the molecular formula of the compound we can calculate percentage composition of the elements. Conversely if we know the percentage composition of the elements initially, we can calculate the relative number of atoms of each element in the molecules of the compound. This gives us the empirical formula of the compound. Further if the molecular mass is known then the molecular formula can be easily determined.

Thus, the **empirical** formula of a compound is a chemical formula showing the relative number of atoms in the simplest ratio, the **molecular formula** gives the actual number of atoms of each element in a molecule.

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i.e. **Empirical formula :** Formula depicting constituent atoms in their simplest ratio.

Molecular formula : Formula depicting actual number of atoms in one molecule of the compound.

The molecular formula is generally an integral multiple of the empirical formula.

i.e. molecular formula = empirical formula × n

where $n = \frac{\text{molecular formula mass}}{\text{empirical formula mass}}$

Example :

Molecular Formula	H_2O_2	C_6H_6	C_2H_6	$C_2H_4O_2$
	2 : 2	6 : 6	2 : 6	2 : 4 : 2
Simplest ratio	1 : 1	1 : 1	1 : 3	1 : 2 : 1
Empirical Formula	H O	C H	CH_3	CH_2O

Determination of Empirical formula:

Following steps are involved in determining the empirical formula of the compounds –

(i) First of all, find the % by wt. of each element present in the compound.

(ii) The % by wt of each element is divided by its atomic weight. It gives atomic ratio of elements present in the compounds.

(iii) Atomic ratio of each element is divided by the minimum value of atomic ratio so as to get simplest ratio of atoms.

(iv) If the value of simplest atomic ratio is fractional then raise the value to the nearest whole number or multiply with suitable coefficient to convert it into nearest whole number

(v) Write the Empirical formula as we get the simplest ratio of atoms.

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Determination of molecular formula:

(i) Find out the empirical formula mass by adding the atomic masses of all the atoms present in the empirical formula of compound.

(ii) Divide the molecular mass (determined experimentally by some suitable method) by the empirical formula mass and find out the value of n.

(iii) Multiply the empirical formula of the compound with 'n' so as to find out the molecular formula of the compound.

Illustration 32

An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73. Molecular formula of compound is :-

Solution

V.D. = 73 $\Rightarrow$ M = 2 × 73 = 146

$C = 146 \times \frac{49.3}{100} = 71.978\ g = \frac{71.978}{12} \simeq 6$ mole

$H = 146 \times \frac{6.84}{100} = 9.9864\ g \simeq 10$ mole

$O = 146 \times \frac{100-(49.3+6.84)}{100} = \frac{64.86}{16} = 64.86\ g \approx 4$ mol

M.F. = $C_6H_{10}O_4$

Illustration 33

The empirical formula of an organic compound containing carbon & hydrogen is CH_2. The mass of 1 litre of organic gas is exactly equal to mass of 1 litre N_2 therefore molecular formula of organic gas is.

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Solution

Empirical Mass of CH_2 = 12 + 2 = 14

$\because$ Mass of 1 litre of organic gas = Mass of 1 litre of N_2

Since V, P, T, n is same.

Therefore, from $PV = \frac{m}{M}RT$ implies that molar mass should also be same.

$\therefore$ Molecular mass of organic compound will be 28 g

$$n = \frac{\text{Molecular mass}}{\text{Empirical mass}} = \frac{28}{14} = 2$$

So molecular formula = 2 × CH_2 = C_2H_4

Illustration 34

In which of following has same empirical formula -

C_2H_2, C_2H_4, C_2H_6, C_6H_6

Solution

C_2H_2, C_6H_6,

Illustration 35

Determine the empirical formula of an oxide of iron, which has 70% Fe and 30% 'O' by mass. (Fe = 56)

Solution

Fe_2O_3

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Stoichiometry:

Stoichiometry is the calculation of amounts of reactants and products involved in a reaction. Stoichiometric calculations require a balanced chemical equation of the reaction.

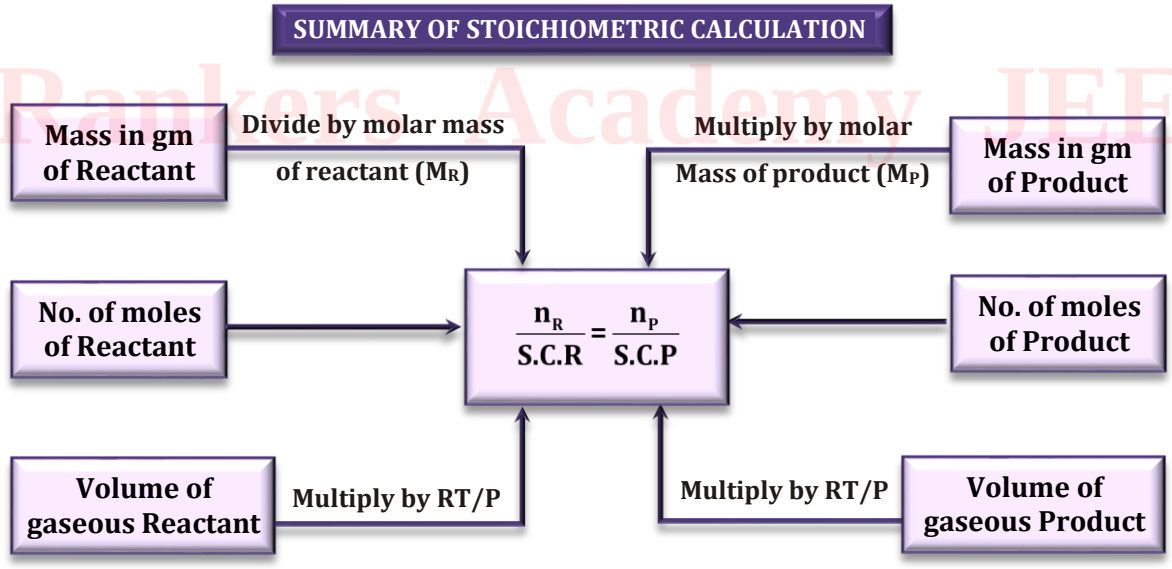
- Remember a balanced chemical equation is one which contains an equal number of atoms of each element on both sides of equation.

Significance of stoichiometric coefficients:

Stoichiometric coefficients of chemical equation tells us about the ratio in which moles of reactants react and moles of products form.

Ex.

	$2H_2(g)$ +	$O_2(g)$ ⟶	$2H_2O(g)$
1st interpretation	2 moles	1 mole	2 moles
2nd interpretation	2 N_A molecules	N_A molecules	2 N_A molecules
3rd interpretation	2 molecules	1 molecules	2 molecules
4rd interpretation (at constant P & T)	2 volume	1 volume	2 volume
5th interpretation (at constant V & T)	2 atm	1 atm	2 atm
6th interpretation	4 g	32 g	36 g



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Here : S.C.R. - Stoichiometric coefficient of reactant,

S.C.P.- Stoichiometric coefficient of product

- Consider a balanced Chemical reaction :

$$a\,A + b\,B \longrightarrow c\,C + d\,D$$

$$\left[\frac{\text{mole of A reacting}}{a} = \frac{\text{mole of B reacting}}{b} = \frac{\text{mole of C produced}}{c} = \frac{\text{mole of D produced}}{d}\right]$$

$\Rightarrow$ If mole of A is n_A, then -

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- mole of B consumed = $n_A \times \left(\frac{b}{a}\right)$
- mole of C produced = $n_A \times \left(\frac{c}{a}\right)$
- mole of D produced = $n_A \times \left(\frac{d}{a}\right)$

I. Mass - mass analysis:

This relates the mass of a species (reactant or product) with the mass of another species (reactant or product) involved in a chemical reaction.

In general, the following steps are adopted in making necessary calculations :

(A) Write down balanced molecular equation for the chemical change.

(B) Write down the relative masses of the reactants and the products with the help of formula below the respective formula. These shall be the theoretical amounts of reactants and product.

(C) Calculate the number of moles below the formula of each of the reactant and product.

(D) By the applications of mole concept, mass co-relation through unitary method or proportionality method the unknown quantity can be determined.

Let us consider a balanced chemical reaction,

$$a\,A + b\,B \longrightarrow c\,C$$

W_A gm ?

If M_A, $M_C \rightarrow$ Molar mass of A and C respectively.

W_A = Mass of A reacted

W_C = Mass of C formed (to be found).

Method $aA + bB \longrightarrow cC$

W_A gm $\qquad\qquad$ $W_C = \left(\frac{c}{a}\right)\frac{W_A}{M_A}M_C$ gm

$\downarrow \qquad\qquad\qquad\qquad \uparrow$

$\left(\frac{W_A}{M_A}\right)$ moles $\rightarrow \qquad\qquad \left(\frac{c}{a}\right)\frac{W_A}{M_A}$ moles

Illustration 36

What mass of CaO is formed by heating 50 g $CaCO_3$ in air ?
[molar mass : $CaCO_3$ = 100, CaO = 56].

Solution

$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

50 gm

$= \frac{50}{100}$ mol

$= \frac{1}{2}$ mol $\quad \frac{1}{2}$ mol

$= \frac{1}{2} \times 56 = 28$ gm

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II. Mass-volume Analysis:

This establishes the relationship between the mass of a species (reactant or product) and the volume of a gaseous species (reactant or product) involved in a chemical reaction.

For example the volume of CO_2 at STP formed by heating 200 g $CaCO_3$:-

$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

200 gm

$= \frac{200}{100}$ mol $= 2$ mol 2 mol

Volume of gas at STP = No. of moles × 22.7 L = 2 × 22.7 = 45.4 L.

III. Volume-volume Analysis:

This relationship deals with the volume of a gaseous species (reactant or product) with the volume of another gaseous species (reactant or product) involved in a chemical reaction.

Gay Lussac's law of combining volumes :

$aA(g) + b(B)(g) \longrightarrow cC(g)$

V_A mL V_B mL V_C mL

reacted reacted formed

$V_A : V_B : V_C = a : b : c$

At constant P, T :

Volume of gaseous species reacting or formed is in ratio of their stoichiometric coefficients.

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Illustration 37

What volume of H_2 (g) is produced by decomposition of 2.4 L NH_3 (g) ?

Solution

$NH_3(g) \longrightarrow \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$

t = 0 2.4 L 0 0

t = ∞ 0 v_1 v_2

Gay-Lussac's law, $2.4 : v_1 : v_2 = 1 : \frac{1}{2} : \frac{3}{2} \Rightarrow \frac{2.4}{v_2} = \frac{2}{3} \Rightarrow v_2 = 3.6$ L

Limiting Reagent (L.R.):

(i) The reactant which is completely consumed when a reaction goes to completion is called Limiting Reactant or Limiting reagent.

(ii) The reactant whose mole ratio is least is limiting reactant.

Where; Mole ratio = $\frac{\text{Given moles of reactant}}{\text{Stoichiometric coefficient of reactant}}$

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(iii) For calculation of moles of product LR should be used. When amounts of more than two reactants are given :

	aA +	bB $\longrightarrow$	cC +	dD
Initial reacting mixture	n_A mol	n_B mol		
mole ratio	$\frac{n_A}{a}$	$\frac{n_B}{b}$		

If $\frac{n_A}{a} < \frac{n_B}{b}$ $\Rightarrow$ A is limiting reagent.

$\frac{n_A}{n_B} = \frac{a}{b}$ then reaction occurs to completion & no reactant is left at the end.

If $\frac{n_A}{n_B} \neq \frac{a}{b}$ then one of reactants is in excess & left at end. The other one which is completely consumed is called **Limiting Reagent (LR)**.

(iv) To find out LR :

	aA +	bB +	cC $\longrightarrow$	dD +	eE
t = 0	n_A mol	n_B mol	n_C mol		

Divide n_A by a, n_B by b & n_C by c, which ever gives smallest no. is LR & all calculations are done using its moles.

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Ex. 1

	$2A$ +	B +	$3C$ $\longrightarrow$	D +	$2E$
t = 0	5 mol	6 mol	6 mol	0	0
	$\frac{5}{2}=2.5$	$\frac{6}{1}=6$	$\frac{6}{3}=2$ (smallest)		

$\therefore$ C is LR

At t = ∞	$5-\frac{12}{3}$ = 1mol	$6-\frac{6}{3}$ = 4 mol	0	2 mol	4 mol

Ex. 2

	$2A(g)$ +	$B(g)$ $\longrightarrow$	$3C(g)$	
At t = 0	2L	3L	0	[$\because$ At constant P and T : $V \propto n$]
	$\frac{2}{2}=1$	$\frac{3}{1}=3$		

$\therefore$ A is LR

At t = ∞	0	$3-\frac{2}{2}=2$L	3L	

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Illustration 38

What weight of AgCl will be precipitated when a solution containing 4.77 g of NaCl is added to a solution of 5.77 g of $AgNO_3$. (Ag = 108)

Ans.

(4.879)

Solution

$AgNO_3$	+	NaCl	$\longrightarrow$	$NaNO_3$	+	AgNCl
5.77 gm				4.77 gm		
0.034 mole (LR)				0.082 mole		

wt. of AgCl = 0.034 × 143.5 = 4.879 gm

Illustration 39

Calculate the mass of sucrose $C_{12}H_{22}O_{11}$ (s) produced by mixing 78 g of C(s), 11 g of H_2(g) & 67.2 litre of O_2 (g) at 0°C and 1 atm according to given reaction (unbalanced)?

$C(s) + H_2(g) + O_2(g) \longrightarrow C_{12}H_{22}O_{11}(s)$

Solution

	$12C(s)$	+	$11\ H_2(g)$	+	$\frac{11}{2}O_2$	$\rightarrow$	$C_{12}H_{22}O_{11}$ (s)
Moles taken	$\frac{78}{12}$				$\frac{11}{2}$		$\frac{67.2}{22.4}$
	= 6.5				= 5.5		= 3
$\frac{\text{moles taken}}{\text{stoich.coeff.}}$	$\frac{6.5}{12}$				$\frac{5.5}{11}$		$\frac{3}{5.5}$
	= 0.54 (L.R.)				= 0.5		= 0.545

$\therefore$ Moles of $C_{12}H_{22}O_{11}$ formed = $\frac{5.5}{11} = 0.5$

Mass of sucrose obtained = 0.5 × 342 = 171 grams.

Illustration 40

How many grams of $KClO_4$ contain 40 gm 'O'.

Solution

$\frac{40}{x} = \frac{64}{35.5+39+64} \Rightarrow x = 86.5$ g

Illustration 41

Calculate mass of Cu in 3.67×10^3 g $CuFeS_2$? (Atomic mass Cu = 63.5, Fe = 56, S = 32)

Solution

Molar mass of $CuFeS_2$ = 183.5

$\because$ $\frac{w_{Cu}}{w_{CuFeS_2}}$ = constant

$\therefore$ $\frac{63.5}{183.5} = \frac{w_{Cu}}{3.67\times10^3} \Rightarrow w_{Cu} = 1270$ g.

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Illustration 42

What mass of H_2SO_4 contains 32 g oxygen ?

Solution

$$\frac{w_O}{w_{H_2SO_4}} = \text{constant} \Rightarrow \frac{64}{98} = \frac{32}{w_{H_2SO_4}} \Rightarrow w_{H_2SO_4} = 49 \text{ g.}$$

Illustration 43

A sample of $MgSO_4$ has 6.022×10^{20} O atoms. What is mass of Mg in sample ?

Solution

In $MgSO_4$; $\frac{\text{mass of O}}{\text{mass of Mg}} = \frac{16\times4}{24} = \frac{8}{3}$ = constant

Mass of 6.022×10^{20} O-atoms $= \frac{6.022\times10^{20}}{6.022\times10^{23}}\times16 = 16 \times 10^{-3}$ g

$\Rightarrow \quad \frac{8}{3} = \frac{16\times10^{-3}}{\text{mass of Mg}} \Rightarrow$ Mass of Mg $= 6 \times 10^{-3}$ g

Illustration 44

If mass % of oxygen in monovalent metal 'M' carbonate (M_2CO_3) is 48%, find atomic mass of metal ?

Solution

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Let A be atomic mass of M

Then $\frac{3\times16}{2A+12+3(16)} \times 100 = 48 \Rightarrow A = 20$

Illustration 45

Find value of X if 36.6 g $BaCl_2.XH_2O$ on strong heating loses 5.4 g moisture ?

[At. mass Ba = 137, Cl = 35.5]

Solution

5.4 g H_2O is in 36.6 g $BaCl_2.XH_2O$

$$\frac{W_{H_2O}}{W_{BaCl_2\times H_2O}} = \text{const.} = \frac{5.4}{36.6} = \frac{18X}{137+71+18X} \Rightarrow X = 2$$

Illustration 46

0.492 g sample of haemoglobin has 0.34% by mass of Fe. If each molecule of haemoglobin has 4 Fe atoms find its molecular mass ?

Solution

Let M = Molecular mass of haemoglobin $\Rightarrow \frac{0.34}{100} \times M = 4 \times 56$

M = 65882.3 a.m.u..

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Principle of Atom Conservation (POAC):

POAC is nothing but the conservation of atoms of reactants and products involved in a chemical reaction, as expressed before in the concepts of atomic theory. And if atoms are conserved, moles of atoms shall also be conserved. The principle is fruitful for the students when they don't get the idea of balanced chemical equation in the problem using POAC we do not need to balance a reaction and we can even add two or more reactions. This principle can be under stood by the following example.

Consider the decomposition of $KClO_3$ (s) $\rightarrow$ KCl (s) + O_2 (g) (unbalanced chemical reaction)

Apply the principle of atom conservation (POAC) for K atoms.

or moles of K atoms in $KClO_3$ = moles of K atoms in KCl

Now, since 1 molecule of $KClO_3$ contains 1 atom of K

Thus, moles of K atoms in $KClO_3$ = 1 × moles of $KClO_3$

and moles of K atoms in KCl = 1 × moles of KCl

$\therefore$ moles of $KClO_3$ = moles of KCl or $\frac{\text{wt. of } KClO_3 \text{ in g}}{\text{mol. wt. of } KClO_3} = \frac{\text{wt. of KCl in g}}{\text{mol. wt. of KCl}}$

- The above equation gives the mass-mass relationship between $KClO_3$ and KCl which is important in stoichiometric calculations. Again, applying the principle of atom conservation for O atoms,

  moles of O in $KClO_3$ = 3 × moles of $KClO_3$

  moles of O in O_2 = 2 × moles of O_2

  $\therefore$ 3 × moles of $KClO_3$ = 2 × moles of O_2

  or $3 \times \frac{\text{wt. of } KClO_3}{\text{mol. wt. of } KClO_3} = 2 \times \frac{\text{vol. of } O_2 \text{ at NTP}}{\text{standard molar vol. (22.4 lt)}}$

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- The above equations thus give the mass-volume relationship of reactants and products.

Illustration 47

27.6 g K_2CO_3 was treated by a series of reagents so as to convert all of its carbon to $K_2Zn_3[Fe(CN)_6]_2$. Calculate the weight of the product. [mol. wt. of K_2CO_3 = 138 and mol. wt. of $K_2Zn_3[Fe(CN)_6]_2$ = 698]

Solution

Here we have no knowledge about series of chemical reactions but we know about initial reactant and final product, accordingly.

$K_2CO_3 \xrightarrow[\text{Steps}]{\text{Several}} K_2Zn_3[Fe(CN)_6]_2$

Since C atoms are conserved, applying POAC for C atoms,

moles of C in K_2CO_3 = moles of C in $K_2Zn_3[Fe(CN)_6]_2$

1 × moles of K_2CO_3 = 12 × moles of $K_2Zn_3[Fe(CN)_6]_2$

($\because$ 1 mole of K_2CO_3 contains 1 moles of C)

$$\frac{\text{wt. of } K_2CO_3}{\text{mol. wt. of } K_2CO_3} = 12 \times \frac{\text{wt. of the product}}{\text{mol. wt. of product}}$$

$$\textbf{wt. of } K_2Zn_3[Fe(CN)_6]_2 = \frac{27.6}{138} \times \frac{698}{12} = \textbf{11.6 g}$$

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Percentage purity:

The percentage of a specified compound or element in an impure sample.

$$\%\,\text{purity} = \frac{\text{Actual mass of compound}}{\text{Total mass of sample}} \times 100$$

Illustration 48

A chalk sample exactly requires 17.52 gram HCl for complete reaction with all $CaCO_3$ present in it. If the chalk sample is 72% pure, the mass of sample taken is

Solution

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

Moles of HCl = $\frac{17.52}{36.5}$

Moles of $CaCO_3$ = $\frac{1}{2} \times \frac{17.52}{36.5}$

Weight of $CaCO_3$ required = $\frac{1}{2} \times \frac{17.52}{36.5} \times 100$

Mass of sample taken :

$= \frac{1}{2} \times \frac{17.52}{36.5} \times \frac{100 \times 100}{72}$ = 33.33 gm

Some Examples:

Illustration 49

For the reaction, $A(s) + 2B(g) \longrightarrow 3C(g) + D(g)$

Molar mass [g mol^{-1}] A = 100; B = 50; C = 60; D =20

(a) How many grams of C are produced by reaction of 250g A ? [Hint : Mass-mass relation]

(b) What mass of B reacts to give 500g D ?

(c) How many grams of A will produce 11.2L of C at STP ?

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Solution

(a) $A(s) + 2B(g) \longrightarrow 3C(g) + D(g)$

$\frac{250}{100}$ =2.5mol excess 7.5mol

= 7.5 × 60g

= 450g

(b) $A(s) + 2B(g) \longrightarrow 3C(g) + D(g)$

50 mol $\frac{500}{20}$ = 25 mol

= 50 × 50

= 2500 g

(c) [Hint : mass-volume relation]

$A(s) + 2B(g) \longrightarrow 3C(g) + D(g)$

$\frac{0.5}{3}$ mol $\frac{11.2}{22.4}$ = 0.5 mol

$= \frac{0.5}{3} \times 100$

= 16.67g

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Illustration 50

18.625g KCl is formed due to decomposition of $KClO_4$ in reaction

$KClO_4(s) \longrightarrow KCl(s) + 2O_2(g)$

Find volume of O_2 obtained at STP. [Atomic mass K = 39, Cl = 35.5]

Solution

$KClO_4(s)$	$\longrightarrow$	$KCl(s)$	+	$2O_2(g)$
		$\frac{18.625}{74.5}$ mol		$\frac{2\times18.626}{74.5}$
		= 0.25		= 0.5 mol

Vol. of O_2 at STP = 0.5 × 22.4 = 11.2 L

Illustration 51

What mass of CH_3OH (methanol) will be produced by reacting 1120 L of $H_2(g)$ at STP with excess CO(g) ?

Solution

Write balanced reaction :

$CO(g)$	+	$2H_2(g)$	$\longrightarrow$	$CH_3OH(\ell)$
$\frac{1120}{22.4}$ mol				25 mol
= 50 mol				

Mass of CH_3OH = 25 × 32 = 800 g

Illustration 52

	$2A(s)$	+	$3B(g)$	$\longrightarrow$	$4C(g)$
t=0	10 mol		10 mol		

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Find final moles of A, B and C.

Solution

LR is B

	2A	+	3B	$\longrightarrow$	4 C
t=∞	$10-10\left(\frac{2}{3}\right)$		0		$10\left(\frac{4}{3}\right)$
t=∞	$\frac{10}{3}$ mol		0		$\left(\frac{40}{3}\right)$ mol

Illustration 53

If 2.4 gm Mg is treated with 0.64 gm O_2. Find composition of final product mixture ?

[Atomic Mass Mg = 24]

Solution

	$Mg(s)$	+	$\frac{1}{2}O_2(g)$	$\rightarrow$	$MgO(s)$
t = 0	0.1 mol		0.02 mol		0
	[O_2 is LR]				
t = ∞	0.1 –0.02 × 2		0		0.02 × 2
	= 0.06 mol		= 0.04 mol		

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Illustration 54

20g of impure NaCl sample is added aqueous solution having excess $AgNO_3$. AgCl precipitate is filtered washed, dried & its mass is 28.7g. Find % purity of NaCl.

[Atomic mass Ag = 108, Cl = 35.5]

Solution

$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(s)$ [Precipitation reaction]

0.2moles $\quad \frac{28.7}{143.5}$ moles

$= 0.2$ moles

Mass of NaCl = $0.2 \times 58.5 = 11.7$ gm

$$\% \text{ purity} = \frac{11.7}{20} \times 100$$

$= 58.8\ \%$

Percentage Yield:

In general, when a reaction is carried out in the laboratory we do not obtain actually the theoretical amount of the product. The amount of the product that is actually obtained is called the actual yield. Knowing the actual yield and theoretical yield the percentage yield can be calculate as :

$$\boxed{\%\text{yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100}$$

Illustration 55

What is mass of O_2 required to produce 960 gm of O_3 if % yield of reaction $3O_2 \rightarrow 2O_3$ is 50% ?

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Solution

$3O_2 \longrightarrow 2O_3$

$\frac{960}{48} = 20\text{mol} = \text{Actual yield}$

Let X mol O_2 be required

$$\text{Theoretical yield of } O_3 = \left(\frac{2}{3}X\right) \Rightarrow 50 = \left(\frac{20}{\frac{2}{3}X}\right)(100)$$

$X = 60$ mol $= 60 \times 32 = 1920$ g O_2 required

Illustration 56

What is mass of C obtained on reacting 20 moles of A with excess B by reaction :

$2A + B \longrightarrow 3C$

If % yield of reaction is 80% ?

Solution

$2A + B \longrightarrow 3C$

20mol $\qquad$ 30 mol = Theoretical yield

Actual yield = X mol = ?

$$\Rightarrow \frac{X}{30} \times 100 = 80 \qquad \Rightarrow X = 24 \text{ moles}$$

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Illustration 57

Aluminium reacts with sulphur to form aluminium sulphide. If 5.4 gm of aluminium reacts with 12.8gm sulphur & gives 12gm of aluminium sulphides, then the percent yield of the reaction is. (Al = 27, S = 32)

Solution

	$2Al$ +	$3S$ $\longrightarrow$	Al_2S_3
Mole taken	$\frac{5.4}{27}$ gm	$\frac{12.8}{32}$ gm	
	= 0.2	= 0.4	
$\frac{\text{moles taken}}{\text{stoich.coeff.}}$	$\frac{0.2}{2}$	$\frac{0.4}{3}$	
	= 0.1 (L.R.)	= 0.133	

moles of Al_2S_3 formed = $\frac{1}{2}\times 0.2 = 0.1$

mass of Al_2S_3 formed = 0.1 × 150 = 15 gm

But, from equation, only 12 gm Al_2S_3 is formed.

% yield = $\frac{\text{actual yield}}{\text{theoritical yield}}\times 100 = \frac{12}{15}\times 100 = 80$ %

Illustration 58

n-butane is product by the monobromination of ethane followed by the Wurtz reaction. Calculate the volume of ethane at NTP required to produce 55g n-butane, if the bromination takes place with 90% yield and the Wurtz reaction with 85% yield.

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Ans. (55.50)

Solution

(i) Monobromination of ethane

$\underset{\text{Ethane}}{C_2H_6} \xrightarrow{Br_2} C_2H_5Br$

(ii) Wurtz reaction :

$2C_2H_5Br + 2Na \xrightarrow[\text{ether}]{\text{Dry}} C_4H_{10} + 2NaBr$

Molecular weight of

$C_4H_{10} = (12 \times 4) + (10 + 1) = 58$

$\therefore$ Amount of n-butane to be produced = $\frac{55}{58}$ mol = 0.948 mol

$\therefore$ Amount of C_2H_5Br required to obtain 0.948 mol

But the conversion is only 85%

Hence, the amount of C_2H_5Br required = $\frac{1.896}{85}$ × 100 = 2.23 mol

To obtain C_2H_5Br from C_2H_6, the same amount of C_2H_6 would be required. But the percent conversion of C_2H_6 to required = $\frac{2.23}{90}\times 100 = 2.478$mol.

Thus, required volume of ethane of NTP = 22400 × 2.478 = 55507.2 ml = 55.50 L.

Problems based on mixture:

The composition of any mixture may be determined by reacting the mixture with some substance, by which either one or more component of mixture may react.

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Illustration 59

1.5 gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighing 1.46 gm. The reaction responsible for loss of weight is

$Fe_2O_3(s) \rightarrow Fe_3O_4(s) + O_2(g)$

What is the percentage by mass of Fe_2O_3 in original sample.

Solution

$3Fe_2O_3(s) \rightarrow 2Fe_3O_4 + \frac{1}{2}O_2$

$3 \times 160 \qquad \frac{1}{2} \times 32$

$= 480 \text{ gm} \qquad = 16 \text{ gm}$

loss of 16 gm $\rightarrow$ 480 gm Fe_2O_3

loss of 0.04 gm $\rightarrow 0.04 \times \frac{480}{16} = 1.2$ gm Fe_2O_3

% by mass $= \frac{1.2}{1.5} \times 100 = 80\%$

Illustration 60

19 gm mixture of $Na_2CO_3(s)$ and $NaHCO_3(s)$ on heating gives 2.2 gm CO_2 gas.

Find % $NaHCO_3$ (by mass) in mixture.

Solution

44.21 %

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Illustration 61

Write decomposition reaction of $CaCO_3(s)$ and Na_2CO_3 (s) on heating.

Solution

$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$

$Na_2CO_3(s) \xrightarrow{\Delta}$ (No decomposition)

Problem Related with Parallel Reaction:

When same two reactants form two or more products by independent reactions.

Illustration 62

Carbon reacts with oxygen forming carbon monoxide and/or carbon dioxide depending an availability of oxygen. Find moles of each product obtained when 160 gm oxygen reacts with (a) 12 g carbon (b) 120 g carbon (c) 72 g carbon.

Solution

(a)

	C	+	$\frac{1}{2}O_2$	$\longrightarrow$	CO	[initially use a reaction using lesser amount of oxygen]
t = 0	1mol		5mol			
t =∞	0		5–0.5	=	1mol	
	(LR)		4.5mol			

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Since CO & O_2 are left, CO_2 will also formed.

	CO +	$\frac{1}{2}O_2$	$\longrightarrow$	CO_2
t = 0	1mol	4.5mol		0
t = ∞	0	4 mol		1 mol

At end, 1 mole CO_2 & no CO present

(b)

	C +	$\frac{1}{2}O_2$	$\longrightarrow$	CO
t = 0	10mol	5mol		0
t =∞	0	0		10mol

At end only 10 mol CO present.

(c)

	C +	$\frac{1}{2}O_2$	$\longrightarrow$	CO
t = 0	6mol	5mol		0
t =∞	0	2mol		6mol
	[LR]			

	CO +	$\frac{1}{2}O_2$	$\longrightarrow$	CO_2
t = 0	6mol	2mol		0
t =∞	2mol	0 [LR]		4mol

At end [2mol CO + 4mol CO_2] left.

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Illustration 63

25.4 gm of iodine and 14.2 gm of chlorine are made to react completely to yield mixture of ICl and ICl_3. Ratio of moles of ICl & ICl_3 formed is (Atomic mass : I = 127, Cl = 35.5)

Solution

I_2 +	Cl_2	$\longrightarrow$	ICl +	ICl_3
0.1mol	0.2 mol		x mol	y mol

Applying conservations of I and Cl -atom

$0.2 = x+y \quad \Rightarrow \quad x = 0.1$

$0.4 = x+3y \quad \Rightarrow \quad y = 0.1$

$\therefore \; n_{ICl} : n_{ICl_3} = x : y = 1 : 1$

Problems Related with Sequential Reaction:

When one of products formed in previous reaction is consumed in the next one.

Illustration 64

How many grams H_2SO_4 can be obtained from 1320 gm PbS as per reaction sequence ?

$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$

$3SO_2 + 2HNO_3 + 2H_2O \longrightarrow 3H_2SO_4 + 2NO$

[At. mass : Pb = 208, S = 32]

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Solution

Moles of PbS = $\frac{1320}{240}$ = 5.5 mol

Moles of SO_2 = 5.5 mol = moles of H_2SO_4

Mass of H_2SO_4 = 5.5 × 98 = 539 gm

[When amount of only one reactant is given generally other is assumed in excess.]

Eudiometry:

Eudiometry or gas analysis involves the calculations based on gaseous reactions or the reactions in which at least two components are gaseous, in which the amounts of gases are represented by their volumes, measured at the same pressure and temperature. Some basic assumptions related with calculations are:

(i) Gay-Lussac's law of volume combination holds good. According to this law, the volumes of gaseous reactants reacted and the volumes of gaseous products formed, all measured at the same temperature and pressure, bear a simple ratio.

$N_2(g)$ + $3H_2(g)$ ⟶ $2NH_3$ (g)

1 vol. 3 vol. 2 vol.

Problem may be solved directly is terms of volume, in place of mole. The stoichiometric coefficients of a balanced chemical reactions gives the ratio of volumes in which gaseous substances are reacting and products are formed, at same temperature and pressure.

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(ii) The volumes of solids or liquids is considered to be negligible in comparison to the volume of gas. It is due to the fact that the volume occupied by any substance in gaseous state is even more than thousand times the volume occupied by the same substance in solid or liquid states.

$2H_2(g)$ + O_2 (g) ⟶ $2H_2O$ (l)

2 mole 1 mole 2 mole

2 vol. 1 vol. 0 vol.

(iii) Air is considered as a mixture of oxygen and nitrogen gases only. It is due to the fact that about 99% volume of air is composed of oxygen and nitrogen gases only.

(iv) Nitrogen gas is considered as a non-reactive gas. It is due to the fact that nitrogen gas reacts only at very high temperature due to its very high thermal stability. Eudiometry is performed in a eudiometer tube and the tube cannot withstand very high temperature. This is why, nitrogen gas cannot participate in the reactions occurring in the eudiometer tube.

(v) The total volume of non-reacting gaseous mixture is equal to sum of partial volumes of the component gases **(Amagat's law)**.

$V = V_1 + V_2 + ...$

Partial volume of gas in a non-reacting gaseous mixture is its volume when the entire pressure of the mixture is supposed to be exerted only by that gas.

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(vi) The volume of gases produced is often given by certain solvent which absorb contain gases.

Solvent	Gases absorb
KOH	CO_2, SO_2, Cl_2
Ammonical Cu_2Cl_2	CO
Turpentine oil	O_3
Alkaline pyrogallol	O_2
water	NH_3, HCl
$CuSO_4/CaCl_2$	H_2O

(vii) EUDIOMETER

A eudiometer is a laboratory device that measures the change in volume of a gas mixture following a physical or chemical change.

Illustration 65

10 ml of CO is mixed with 25 ml air (20% O_2 by volume). Find final volume (in ml) after complete combustion.

Solution

$$\overset{10ml}{CO} + \frac{1}{2}\overset{5ml}{O_2} \longrightarrow \underset{10ml}{CO_2}$$

$V_f = V_{CO_2}$ + Volume of remaining air = 10 + 20 = 30 ml

Illustration 66

A 3 L gas mixture of propane (C_3H_8) and butane (C_4H_{10}) on complete combustion at 25°C produced 10 L CO_2. Assuming constant P and T conditions what was volume of butane present in initial mixture ?

Solution

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$C_3H_8(g) + 5O_2 \longrightarrow 3CO_2(g) + 4H_2O(l)$

x L 3x L

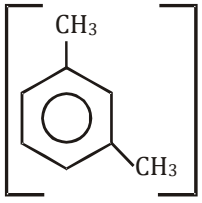
$C_4H_{10}(g) + \frac{13}{2}O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$

(3-x) L 4(3 – x) L

from question $3x + 4(3 - x) = 10 \Rightarrow x = 2$

$\therefore$ Volume of butane, $C_4H_{10} = (3 - x) = 1$ L

Illustration 67

100 ml gaseous meta Xylene  undergoes combustion with excess of oxygen at room temperature and pressure. Volume contraction / expansion (in ml) during reaction is

Solution

$$C_8H_{10}(g) + \frac{21}{2}O_2(g) \longrightarrow 8CO_2(g) + 5H_2O(l)$$

100 ml $\frac{21}{2}\times 100 = 1050ml$ 800 ml 0

$\therefore$ Contraction in volume = (100 + 1050) – 800 = 350 ml

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Illustration 68

An alkene upon combustion produces $CO_2(g)$ and $H_2O(g)$. In this combustion process if there is no volume change occurs then the no. of C atoms per molecule of alkene will be :

Solution

$$C_nH_{2n}(g) + \frac{3n}{2} O_2(g) \rightarrow nCO_2(g) + nH_2O(g)$$

if there no volume changes i.e. $\Delta_{ng} = 0$

$$(n + n) - \left(1 + \frac{3n}{2}\right) = 0 \Rightarrow n = 2$$

Illustration 69

A gaseous hydrocarbon (C_xH_y) requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times of its volume of CO_2. Find out the volume of x + y.

Solution

$$C_XH_y + (x + \frac{y}{4}) O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O(l)$$

a $\quad a\left(x + \frac{y}{4}\right)$ $\quad ax$

Given that : $a(x + y/4) = 6a$

and $\quad ax = 4\ (a)$

$\quad x = 4$ and $y = 8$

$\therefore \quad x + y = 4 + 8 = 12$

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Illustration 70

On heating 60 ml mixture containing equal volume of chlorine gas and it's gaseous oxide, volume becomes 75 ml due complete decomposition of oxide. On treatment with KOH volume becomes 15 ml. What is the formula of oxide of chlorine ?

Solution

Let oxide of Cl is Cl_xO_y

So, in 60 mL $\Rightarrow$ 30 mL Cl_xO_y and 30 mL Cl_2.

Now,

$$Cl_xO_y \longrightarrow \frac{x}{2}Cl_2 + \frac{y}{2}O_2$$

30mL

$\qquad \frac{30.x}{2}$ mL $\qquad \frac{30.y}{2}$ mL

Given :

$$75 = 30 + \frac{30x}{2} + \frac{30y}{2} \Rightarrow x + y = 3 \qquad \text{...(i)}$$

KOH absorbs Cl_2 and volume becomes 15 mL so,

$$(75 - 15) = V_{Cl_2} = 30 + \frac{30x}{2} \Rightarrow x = 2 \text{ and } y = 1$$

So the oxide : Cl_2O

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Illustration 71

5 L of A (g) & 3 L of B(g) measured at same T & P are mixed together which react as follows

$$2A(g) + B(g) \rightarrow C(g)$$

What will be the total volume (in litre) after the completion of the reaction at same T & P.

Solution

$2A(g) + B(g) \longrightarrow C(g)$

5L 3L

L.R. is A

So, volume of C produced = $\frac{1}{2}\times 5 = 2.5$ L

and, volume of B reacted = $\frac{1}{2}\times 5 = 2.5$ L

So, volume of B remained = 3 – 2.5 = 0.5 L

Hence, $V_{total} = V_C + V_B = 2.5 + 0.5 = 3$ L

Solutions:

A solution is a homogenous mixture of two or more pure substances whose composition may be altered within certain limits. Though the solution is homogenous in nature, yet it retains the properties of its constituents.

Generally, solution is composed of two components, **solute** and **solvent**. Such type of solution is known as binary solution. Solvent is that component in solution whose physical state is the same as that of the resulting solution while another component is called as solute. If the physical state of both components is same, then the component in excess is known as solvent and other one is called as solute. Each component in a binary solution can be in any physical state such as liquid, solid and gaseous state.

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Types of Solutions

Type of Solutions	Solute	Solvent	Common Example
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

Concentration or strength of solution :

The concentration of a solution is the amount of solute dissolved in a known amount of the solvent or solution. Solution can be described as dilute or concentrated solution as per their concentration. A dilute solution has a very small quantity of solute while concentrated solution has a large quantity of solute in solution. Various concentration terms are as follows.

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Mass percentage (w/W %):

It may be defined as the number of parts of mass of solute per hundred parts by mass of solution.

$$\% \text{ by mass } \left(\frac{w}{W}\right): = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100$$

[X % by mass means 100 g solution contains X g solute and hence (100 – X) g solvent]

Mass-volume percentage (w/V %) :

It may be defined as the mass of solute (in gm) present in 100 cm^3 of solution. For example, if 100 cm^3 of solution contains 5 g of sodium hydroxide, then the mass-volume percentage will be 5% solution.

$$\% \left(\frac{w}{V}\right) = \frac{\text{wt. of solute}}{\text{volume of solution}} \times 100$$

[X % $\left(\frac{w}{V}\right)$ means 100 ml solution contains X gm solute]

Volume Percent (v/V %):

It can be represented as % v/v or % volume and normally used for the solutions in which both components are in liquids state. It is the number of parts of by volume solute per hundred parts by volume of solution. Therefore,

$$\% \left(\frac{v}{V}\right) = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100$$

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Parts per million (ppm) :

The very low concentration of solute in solution can be expressed in ppm. It is the number of parts by mass of solute per million parts by mass of the solution.

$$\text{Parts per million (ppm)} = \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6 \cong \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

Get yourselves very much comfortable in their inter conversion. It is very handy.

Molarity (M) :

Molarity is most common unit for concentration of solution. It is defined as the number of moles of solute present in one litre or one dm^3 of the solution or millimole of solute present in one mL of solution.

$$\text{Molarity (M)} = \frac{\text{Mole of solute}}{\text{volume of solution in litre}}$$

Molality (m) :

The number of mole of the solute present in 1000 g of the solvent is known as molality of solution. It represented by letter 'm'.

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

The unit of molality is mol/kg and it does not affect by temperature.

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Mole Fraction (X) :

Mole fraction may be defined as the ratio of number of moles of one component to the total number of moles of all the components (solute and solvent) present in solution. It is denoted by letter X and the sum of all mole fractions in a solution is always equals to one.

$$\text{Mole fraction (X)} = \frac{\text{Moles of solute}}{\text{Total moles}}$$

Mole fraction does not depend upon temperature and can be extended to solutions having more than two components.

Concentration Type	Mathematical Formula	Concept
Percentage by mass	$\%\left(\frac{w}{w}\right) = \frac{\text{Mass of solute} \times 100}{\text{Mass of solution}}$	Mass (in g) of solute present in 100 gram of solution.
Volume percentage	$\%\left(\frac{v}{v}\right) = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$	Volume (cm^3) of solute present in 100 cm^3 of solution.
Mass-volume percentage	$\%\left(\frac{w}{v}\right) = \frac{\text{Mass of solute} \times 100}{\text{Volume of solution}}$	Mass (in g) of solute present in 100 cm^3 of solution.
Parts per million	$\text{ppm} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$	Parts by mass of solute per million parts by mass of the solution
Mole fraction	$X_A = \frac{\text{Mole of A}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + ...}$ $X_B = \frac{\text{Mole of B}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + ...}$	Ratio of number of moles of one component to the total number of moles.
Molarity	$M = \frac{\text{Mole of solute}}{\text{Volume of solution (in L)}}$	Moles of solute in one litre of solution.
Molality	$m = \frac{\text{Mass of solute} \times 1000}{\text{Molar mass of solute} \times \text{Mass of solvent(g)}}$	Moles of solute in one kg of solvent.

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Illustration 72

Calculate the mole fractions of the components of the solution composed by 92 g glycerol and 90 g water ? (M (water) = 18; M (glycerol) = 92)

Solution

Moles of water = 90 g/18 g = 5 mol water

Moles of glycerol = 92 g/92 g = 1 mol glycerol

Total moles in solution = 5 + 1 = 6 mol

Mole fraction of water = 5 mol/6 mol = 0.833

Mole fraction of glycerol = 1 mol/6 mol = 0.167

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Illustration 73

What will be the molarity of solution when water is added to 16.4g $Ca(NO_3)_2$ to make 100 mL of solution?

Solution

Mol of $Ca(NO_3)_2$ = 16.4/164 = 0.1

Molarity = Mole of solute/Volume of solution (L) = 0.10 mol/0.10 L

Therefore, Molarity of given solution = 1.0 M

Illustration 74

Calculate the molality of a solution containing 20 g of sodium hydroxide (NaOH) in 250 g of water?

Solution

Moles of sodium hydroxide = 20/40 = 0.5 mol NaOH

250 g = 0.25 kg of water

Hence molality of solution = Mole of solute/Mass of solvent (kg)= 0.5 mol/0.25 kg

or Molality(m) = 2.0 m

Illustration 75

Calculate the gram of copper sulphate ($CuSO_4$) needed to prepare 250.0 mL of 1.00 M $CuSO_4$?

Solution

Moles of $CuSO_4$ = M × V = $1 \times \frac{250}{1000}$

Molar mass of copper sulphate = 159.6 g/mol

Hence Mass of copper sulphate (g) = Moles of $CuSO_4$ × Molar mass of copper sulphate.

$= 1 \times \frac{250}{1000} \times 159.6$ g/mol = 39.9 g of Copper sulphate

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Illustration 76

How many gram of H_2SO_4 are present in 500 ml of 0.2M H_2SO_4 solution ?

Solution

$M = \frac{\text{moles}}{\text{vol.}} \Rightarrow$ moles of H_2SO_4 = M × V = $0.2 \times \frac{500}{1000}$ L = 0.1

Mass of H_2SO_4 = 0.1 × 98 = 9.8 g

Illustration 77

Calculate the ppm of mercury in water in a sample containing 30 mg of Hg in 500 ml of solution.

Solution

$\text{Parts per million} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$

Mass of Hg = 30 mg

Mass of water = 500/1 = 500g = 50×10^4 mg

(density = mass/volume; density of water 1 g/ml) $w = \frac{v}{d}$

Therefore, ppm of mercury = $\frac{30 \times 10^6}{50 \times 10^4}$ = 60 ppm of mercury

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Illustration 78

A 100g NaOH solution has 20g NaOH. Find molality.

Solution

$$m = \frac{20/40}{100-20} \times 1000 = \frac{500}{80} = 6.25 \text{mol/kg}$$

Illustration 79

Find molality of aqueous solution of CH_3COOH whose molarity is 2M and density d = 1.2 g/mL.

Hint : $\frac{1000 \times M}{1000 \times d - MMs}$

where d = density in gL^{-1}, M = Molarity, m = molality, M_S = molar mass of solute.

Solution

$$m = \frac{2}{1200-2\times60} \times 1000 = 1.85 \text{ m}$$

Illustration 80

A solution has 80% $\frac{w}{w}$ NaOH with density $2gml^{-1}$. Find (a) Molarity, (b) Molality of solution.

Solution

Let V be vol. of solution, in L

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$$\text{Mass of solute} = (d \times V) \times \frac{\left(\%\frac{w}{w}\right)}{100} = 2 \times V \times \frac{80}{100} = 1.6V$$

(a) $M = \frac{1.6V/40}{V} = 0.04$ m

(b) $m = \frac{1.6V/40}{2V-1.6V} \times 1000 = 100 \text{ mol kg}^{-1}$

Illustration 81

4.450 g sulphuric acid was added to 82.20 g water and the density of the solution was found to be 1.029 g/cc at 25°C and 1 atm pressure. Calculate

(a) the weight percent,

(b) the mole fraction,

(c) the mole percent,

(d) the molality,

(e) the molarity of sulphuric acid in the solution under these conditions.

Solution

Sulphuric acid = 4.450 g, Water = 82.20 g $\Rightarrow$ Wt. of solution = 86.65 g

$\therefore$ Density of solution = 1.029 g/cc.

(a) Weight percent = $\frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100 = \frac{4.450}{86.65} \times 100 = 5.14$

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(b) Mole fraction :

$$\text{Mole of solute} = \frac{\text{wt. of solute}}{\text{mol wt. of solute}} = \frac{4.45}{98} = 0.0454$$

$$\text{Mole of solvent} = \frac{82.20}{18} = 4.566$$

Total moles in solution = 0.0454 + 4.566 = 4.6114

$$\text{Mole fraction of solute} = \frac{0.0454}{4.6114} = 0.0098$$

(c) Mole percent = mole fraction of solute × 100 = 0.0098 × 100 = 0.98

(d) $$\text{Molality} = \frac{\text{moles of solute}}{\text{mass of solvent (in gm)}} \times 1000$$

$$= \frac{0.0454 \times 1000}{82.2} = 0.552$$

(e) $$\text{Molarity} = \frac{\text{moles of solute}}{\text{litre of solution}}$$

$$\text{Volume of solution} = \frac{\text{Mass}}{\text{Density}} = \frac{86.65}{1.029}\text{ml} = \frac{86.65}{1.029 \times 1000}\text{litre}$$

$$\text{Molarity} = \frac{0.0454}{\frac{86.54}{1.029 \times 1000}} = \frac{0.0454 \times 1000 \times 1.029}{86.65} = 0.539$$

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Illustration 82

Find number of $Na^{\oplus}$ & $PO_4^{3\ominus}$ ions in 250 ml of 0.2M Na_3PO_4 solution.

Solution

Na_3PO_4 + aq. $\longrightarrow$ $3Na^{\oplus}(aq) + PO_4^{3\ominus}(aq)$ [Ionic compound when added to water ionize completely].

50 millimoles (m.m.) 150 mm 50 mm

No. of $Na^{\oplus}$ ions = $150 \times 10^{-3} \times N_A$; No. of $PO_4^{3\ominus}$ ions = $50 \times 10^{-3} \times N_A$

Illustration 83

80g NaOH was added to 2L water. Find molality of solution if density of water = 1g/mL

Solution

$$m = \frac{\text{moles of NaOH}}{\text{mass of } H_2O} \times 1000 = \frac{80/40}{2 \times 1000} \times 1000 = 1 \text{ molal}$$

Illustration 84

The average concentration of $Na^{\oplus}$ ion in human body is 3.0 to 3.9 gm per litre. The molarity of $Na^{\oplus}$ ion is about.

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Solution

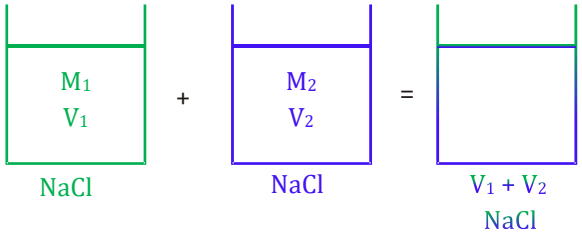
0.15 M

$$M_{Na^{\oplus}} = \frac{n_{solute}}{\text{volume of solution in Lt}} = \frac{\frac{3+3.9}{2}}{23} = \frac{6.9}{46} = 0.15\ M$$

Mixing of and Dilution Solutions:

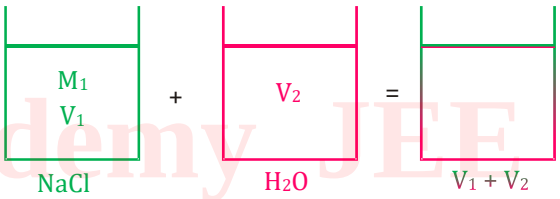
It is based on law of conservation of moles of solute

(i) Two solutions having same solute



$$\text{Final molarity} = \frac{\text{Total moles}}{\text{Total volume}} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

(ii) Dilution Effect : When a solution is diluted, the moles of solute do not change but molarity changes while on taking out a small volume of solution from a larger volume, the molarity of solution do not change but moles change proportionately.



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$$\text{Final molarity} = \frac{M_1V_1}{V_1 + V_2}$$

n-fold or n-times dilution

⇒ Final volume = $V_1 + V_2 = n(V_1)$

Illustration 85

50 ml 0.2 M H_2SO_4 is mixed with 50 ml 0.3M H_2SO_4. Find molarity of final solution.

Solution

$$M_f = \frac{\text{Total moles of } H_2SO_4}{\text{Total volume}} = \frac{50\times0.2\times10^{-3} + 50\times10^{-3}\times0.3}{(50+50)\times10^{-3}} = 0.25\ M.$$

Illustration 86

Find final molarity in each case :

(i) 500 ml 0.1 M, HCl + 500 ml 0.2M HCl

(ii) 50 ml, 0.1M HCl + 150 ml, 0.3MHCl + 300 ml H_2O

(iii) 4.9g H_2SO_4 + 250 ml, H_2O + 250 ml 0.1 M H_2SO_4

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Solution

(i) $M_f = \frac{500\times0.1+500\times0.2}{500+500} = 0.15$ M.

(ii) $M_f = \frac{50\times0.1+150\times0.3}{50+150+300} = \frac{50}{500} = 0.1$ M

(iii) $M_f = \frac{\frac{4.9}{98}+\frac{250}{1000}\times0.1}{\left(\frac{250+250}{1000}\right)} = \frac{50+25}{500} = 0.15$ M

Illustration 87

How much water should be added to 2M HCl solution to form 1 litre of 0.5 M HCl ?

Solution

Let V be initial volume

Then mol of HCl = constant

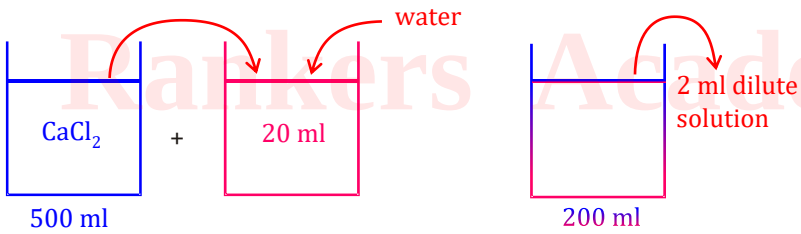
$2 \times V = 1 \times 0.5 \Rightarrow V = 0.25$ L

Volume of water added = 1 – 0.25 = 0.75 L

Illustration 88

1.11g $CaCl_2$ is added to water forming 500 ml of solution. 20 ml of this solution is taken and diluted 10 folds. Find moles of $Cl^{\ominus}$ ions in 2 ml of diluted solution.

Solution



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$\frac{1.11}{111} = 0.01$ mol $CaCl_2$

Moles of $CaCl_2$ in 20ml solution = $\frac{0.01}{500} \times 20 = \frac{0.01}{25}$

In 200 ml solution, moles of $CaCl_2 = \frac{0.01}{25}$ [**Note :** Dilution does not change moles of solute]

In 2 ml of dilute solution moles of $CaCl_2 = \frac{0.01/25}{200} \times 2 = \frac{0.01}{2500} = 4 \times 10^{-6}$

$\therefore$ moles of $Cl^{\ominus} = 2 \times 4 \times 10^{-6} = 8 \times 10^{-6}$

Illustration 89

What volumes of 1M & 2M H_2SO_4 solution are required to produce 2L of 1.75M H_2SO_4 solution?

Solution

Let XL be vol. of 1M solution.

$\therefore$ (2 – X)L is vol. of 2M solution.

Moles of H_2SO_4 : $2 \times 1.75 = 1(X) + (2 - X)2 \Rightarrow X = 0.5$ L

i.e. 0.5L of 1M & 1.5 L of 2M solution are required.

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Illustration 90

A solution is made by mixing 300 ml 1.5M $Al_2(SO_4)_3$ + 300 ml 2M $CaSO_4$ + 400 ml 3.5M $CaCl_2$.

Find final molarity of (1) $SO_4^{2\ominus}$, (2) $Ca^{2\oplus}$, (3) $Cl^{\ominus}$. [Assume complete dissociation of these compounds].

Solution

(1) $[SO_4^{2\ominus}]_f = \frac{\text{Total moles}}{\text{Total volume}} = \frac{300\times1.5\times10^{-3}\times3+300\times2\times10^{-3}}{(300+300+400)\times10^{-3}} = 1.95M$

(2) $[Ca^{2\oplus}]_f = \frac{300\times2+400\times3.5}{1000} = 2M$

(3) $[Cl^{\ominus}]_f = \frac{400\times3.5\times2}{1000} = 2.8M$

Illustration 91

A solution of KCl has a density of 1.69 g mL^{-1} and is 67% by weight. Find the density of the solution if it is diluted so that the percentage by weight of KCl in the diluted solution is 30%.

Solution

Let the volume of the KCl solution be 100 mL,

Weight of KCl solution = 100 × 1.69 = 169 g

100 g of solution contains = 67 g of KCl

169 g of solution = $\frac{67}{100}\times169 = 113.23g$

Lex x mL of H_2O be added.

New volume of solution = (100 + x) mL

New weight of solution = (169 + x) g

(Since x mL of H_2O = x g of H_2O, $d_{H_2O} = 1$)

New percentage of the solution = 30%

$\%\text{ by weight} = \frac{\text{weight of solute}\times100}{\text{weight of solution}}$

$30 = \frac{113.23}{(169+x)}\times100$

x = 208.43 mL = 208.43 g

$\text{New density} = \frac{\text{New weight of solution}}{\text{New volume of solution}} = \frac{(169+x)}{(100+x)}$

$\therefore$ d = 1.224 g/m*l*

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Illustration 92

Calculate the amount of the water "in mℓ" which must be added to a given solution of concentration of 40 mg silver nitrate per ml, to yield a solution of concentration of 16 mg silver nitrate per ml ?

Solution

Before dilution		After dilution
(n_{solute})	=	(n_{solute})
$M_i V_i$	=	$M_f V_f$
$\frac{\frac{40}{170}}{1}\times 1$	=	$\frac{\frac{16}{170}}{1}(1+V)$

$\Rightarrow$ V = 1.5 mL

Illustration 93

100 m*l*, 3%(w/v) NaOH solution is mixed with 100 m*l*, 9%(w/v) NaOH solution. The molarity of final solution is-

Ans. (1.5)

Solution

Total NaOH in 100 ml (1st solution) = 3 g

Total NaOH in 100 ml (2nd solution) = 9 g

$\therefore$ Molarity = $\left(\frac{12/40}{200/1000}\right)=1.5M$

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Illustration 94

1120 gm of 2 **'m'** urea solution is mixed with 2480 gm of 4 **'m'** urea solution. Calculate the molality of the resulting solution?

Solution

3.33 m

Let 2 m, 1120 g solution have mass of solute = w g

$\therefore$ solvent = (1120 – w) g

& Let 4 m, 2480 g solution have mass of solute = w^1 g

$\therefore$ solvent = (2480 – w) g

$$\text{molality} = \frac{n_{solute}}{\text{wt. of solvent in kg}}$$

$$2 = \frac{w/60}{1120-w}\times 1000$$

w = 120 g

$$\& \quad 4 = \frac{w'/60}{2480-w'}\times 1000$$

w' = 480 g

$$\text{resulting molality m} = \frac{\frac{120+480}{60}}{1120-120+2480-480}\times 1000 = 3.33 \text{ m}$$

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Volume Strength of H_2O_2 Solution:

Labelled as 'volume H_2O_2, it means volume of O_2 (in litre) at STP that can be obtained from 1 litre of H_2O_2 solution, when H_2O_2 when it decomposes according to

$$H_2O_2 \rightarrow H_2O + \frac{1}{2} O_2$$

Volume Strength of H_2O_2 Solution = 11.35 × molarity

Illustration 95

Find the % w/v of "10 V" H_2O_2 solution-

Solution

$$\text{Molarity (M) of solution} = \frac{\text{volume strength}}{11.35} = \frac{10}{11.35}$$

$$\%\left(\frac{w}{v}\right) = \frac{M \times \text{mol. wt. of solute}}{10} = \frac{10}{11.35} \times \frac{34}{10} = 3\%$$

Illustration 96

$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

Under conditions where 1 mole of gas occupies 24 dm^3, X L of $\frac{1}{24}$ M solution of H_2O_2 produces 3 dm^3 of O_2.

Thus X is :-

Ans. (6)

Solution

$$\text{moles of } H_2O_2 = \frac{1}{24} \times X$$

$$\text{moles of } O_2 = \frac{3}{24} = \frac{1}{8}.$$

$$\text{moles of } H_2O_2 = \frac{1}{4} = \frac{X}{24}$$

$X = 6$

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Normality:

Normality of solution is defined as the number of equivalents of solute present in one litre (1000 mL) of solution. Let a solution is prepared by dissolving W g of solute of eq. wt. 'E' in 'V' mL water.

- No. of equivalent of solute = $\frac{W}{E}$
- V mL of solution have $\frac{W}{E}$ equivalent of solute
- 1000 mL solution have $\frac{W \times 1000}{E \times V(mL)}$
- **Normality (N) =** $\frac{W \times 1000}{E \times V(mL)}$
- **Normality (N) = Molarity × Valence factor**
  Normality (N) = Molarity (M) × Valence factor (n)
  or N × V (in mL) = M × V (in mL) × n
  or milli equivalents = millimoles × n

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Illustration 97

What volume of 6 M HCl and 2 M HCl should be mixed to get two litre of 3 M HCl ?

Solution

Let, the volume of 6 M HCl required to obtain 2 L of 3M HCl = x L

∴ Volume of 2 M HCl required = (2 – x) L

M_1V_1 + M_2V_2 = M_3V_3

6M HCl 2M HCl 3M HCl

$6 \times (x) + 2 \times (2 - x) = 3 \times 2$

$\Rightarrow$ $6x + 4 - 6x = 6 \Rightarrow 4x = 2$

∴ $x = 0.5$ L

Hence, volume of 6 M HCl required = 0.5 L

Volume of 2M HCl required = 1.5 L

Illustration 98

In a reaction vessel, 1.184 g of NaOH is required to be added for completing the reaction. How many millilitre of 0.15 M NaOH should be added for this requirement ?

Solution

Amount of NaOH present in 1000 mL of 0.15 M NaOH = 0.15 × 40 = 6 g

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∴ 1 mL of this solution contain NaOH = $\frac{6}{1000} \times 10^{-3}$ g

∴ 1.184 g of NaOH will be present in = $\frac{1}{6 \times 10^{-3}} \times 1.184 = 197.33$ mL

Illustration 99

What weight of Na_2CO_3 of 85% purity would be required to prepare 45.6 mL of 0.235N H_2SO_4.

Solution

Meq. of Na_2CO_3 = Meq. of H_2SO_4 = 45.6 × 0.235

∴ $\frac{W_{Na_2CO_3}}{E_{Na_2CO_3}} \times 1000 = 45.6 \times 0.235$

$\Rightarrow$ $\frac{W_{Na_2CO_3}}{106/2} \times 1000 = 45.6 \times 0.235$

∴ $W_{Na_2CO_3} = 0.5679$ g

For 85 g of pure Na_2CO_3, weighed sample = 100 g

∴ For 0.5679 g of pure Na_2CO_3, weighed sample = $\frac{100}{85} \times 0.5679 = 0.6681$ g

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